
QUALITY ASSURANCE PROJECT PLAN
Development and Implementation of a Water Quality Monitoring
Project in Shallow Coral Reef Areas around Puerto Rico
2022-2024



University of Puerto Rico
Mayaguez Campus
Department of Marine Sciences
Caribbean Coral Reef Institute



QUALITY ASSURANCE PROJECT PLAN
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Shallow Coral Reef Areas around Puerto Rico
2022-2024

Revised September~~May~~ 202~~3~~2



UNIVERSITY OF PUERTO RICO - MAYAGUEZ CAMPUS
DEPARTMENT OF MARINE SCIENCES
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CARIBBEAN CORAL REEF INSTITUTE
Dr. Juan José Cruz Motta, Director
Dr. Travis Courtney, Researcher

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TITLE AND APPROVED PAGE

QUALITY ASSURANCE PROJECT PLAN
Development and Implementation of a Water Quality Monitoring Project in Shallow Coral
Reef Areas around Puerto Rico
2022-2024

~~July~~September 2023~~2~~

Department of Natural and Environmental Resources

Responsible Agency

Responsible Person

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**QAPP Development and Implementation of a Water Quality
Monitoring Project in Shallow Coral Reef Areas around Puerto Rico**

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DNER Water Quality Area Sub-Manager

Date:

Wanda García Hernández

**QAPP Development and Implementation of a Water Quality
Monitoring Project in Shallow Coral Reef Areas around Puerto Rico**

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ACRONYMS AND ABBREVIATIONS:

BCG	Biological Condition Gradient
BOD	Biochemical Oxygen Demand
CCRI	Caribbean Coral Reef Institute
CO ₂	Carbon Dioxide
COC	Chain of Custody
CWA	Clean Water Act
DMS	UPR Department of Marine Science
DIC	Dissolved Inorganic Carbon
DO	Dissolved Oxygen
DOD	Diving Officer
DNER	Department of Natural and Environmental Resources
DQO	Data Quality Objective
EPA	United States Environmental Protection Agency
GPS	Global Positioning System
IM	Information Management

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MDL	Method Detection Limit
NEC	Net Ecosystem Calcification
NOAA	National Oceanographic and Atmospheric Administration
pH	Potential of Hydrogen
PRCRM P	Puerto Rico Coral Reef Monitoring Program
QA	Quality Assurance
QAPP	Quality Assurance Project Plan
QAM	Quality Assurance Manager
QA/QC	Quality Assurance/ Quality Control
SM	Standard Method
SOP	Standard Operating Procedure
SAP	Sampling and Analysis Plan
SCUBA	Self-contained underwater breathing apparatus
TA	Total Alkalinity
TSS	Total Suspended Solids
TKN	Total Kjeldahl Nitrogen
TN	Total Nitrogen
TP	Total Phosphorus
UPR	University of Puerto Rico
USVI	U.S. Virgin Islands
WQX	Water Quality Data Portal

MEASUREMENTS UNITS ABBREVIATIONS:

cfs		cubic feet per second
cfu		colony forming unit

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mg/kg		milligram per kilogram
mg/L		milligram per liter
mg/m ³		milligram per cubic meter
mi ²		square mile
mL		milliliter
µg/kg		microgram per kilogram
µg/L		microgram per liter
µS/cm		microsiemens per centimeter
ng		nanogram
ppb		parts per billion
ppm		parts per million
SU		standard units
µmol/kg		micromoles per kilogram

**QAPP Development and Implementation of a Water Quality
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FOREWORD

This Quality Assurance Program Plan (QAPP) pertains to water quality and coral demographics data collection by the University of Puerto Rico (UPR), Department of Marine Sciences (DMS), Caribbean Coral Reef Institute (CCRI). It addresses all chemical, physical and biological monitoring to be performed by CCRI from 2022 through 2024. Appendices as part of this QAPP include field and laboratory Standard Operating Procedures (SOPs) and other supporting documentation.

For additional information that is not contained in this QAPP, see other applicable and current CCRI policies, procedures and plans.

CCRI's programmatic QAPP is generally consistent with the intent of US EPA's Quality Policies (<https://www.epa.gov/quality>) and US EPA-Region 2 Quality Assurance Policies (<https://www.epa.gov/quality/region-2-quality-assurance-guidance-and-standard-operating-procedures>)

US EPA guidance and requirement documents used to guide development of this QAPP include:

- EPA Guidance for Quality Assurance Project Plans (QA/G-5; EPA/240/R-02/009, December 2002)
- EPA Region 2 Guidance for the Development of Quality Assurance Project Plans for Environmental Monitoring Projects, April 12, 2004
- EPA Guidance on Systematic Planning using the Data Quality Objectives Process (QA/G-4; EPA/240/B-06/001, February 2006)
- EPA Guidance for Standard Operating Procedures (QA/G-6, EPA/600/B-07/001; April 2007)
- EPA Guidance on Environmental Data Verification and Data Validation (QA/G-8, EPA/240/R-02/004; November 2002 and reissued January 2008)

Document Availability

The QAPP (main report and appendices) will be made available electronically at CCRI's website: <https://www.uprm.edu/ccri/>~~<http://ccri.uprm.edu>~~

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SECTION A: PROJECT MANAGEMENT

A1. DISTRIBUTION LIST

A copy of this plan will be distributed to all personnel who will participate in the collection, analysis, review, and validation of environmental data for the Water Quality Monitoring Project in Shallow Coral Reef Areas around Puerto Rico Project and those who are responsible for managerial or quality assurance aspects of the project. Copies will also be made available to anyone interested in the quality program for this study.

UPRM/DMS/CCRI in

Dr. Juan José Cruz Motta
Dr. Ernesto Weil
Dr. Travis Courtney
Chemistry Lab Manager
Field Operations Manager
Field Research Assistants 1 & 2
Student Assistant

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Subcontracted personnel:

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Eng. Angel Melendez Aguilar
Wanda Garcia Hernandez, WQA Sub-Manager
Annette Feliberty Ruiz
Angela Garcia Fernandez, Project Manager
Frances M. Segarra Román, Quality Assurance Manager

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USEPA

Izabela Wojtenko
Carol Lynes, USEPA Quality Assurance Officer

A2. PROJECT DESCRIPTION & ORGANIZATION

This QAPP covers water quality monitoring at Puerto Rico's Coral Reef Monitoring Program (PRCRMP) shallow water sites (coral reef areas located at depths less than or equal to 30 meters). Additionally, the QAPP covers coral demographic monitoring at PRCRMP sites and implementation of the Biological Condition Gradient (BCG) from PRCRMP and coral demographic data.

This project is administered and conducted out of the University of Puerto Rico (UPR), Department of Marine Sciences (DMS), Caribbean Coral Reef Institute (CCRI) in La Parguera, PR. CCRI conducts scientific research and monitoring programs that address short and long-term

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management-driven priorities for the understanding and managing of the coral reef ecosystems of the U.S. Caribbean. These activities address the major threats to coral reefs within an ecosystem context, including human socio-economic dimensions, and enhance understanding of the biological and physical processes that structure coral reefs and impact the health of the ecosystem. A total of 42 coral reef sites will be visited 4 times a year for 16 water quality parameters resulting in an annual total of 2,688 data points per year.

Figure 1 provides an overview of specific personnel involved in data collection, analysis, and assessment by CCRI as well as DNER personnel. **Table 1** provides more detailed descriptions of the roles and responsibilities for these CCRI staff and subcontractors.

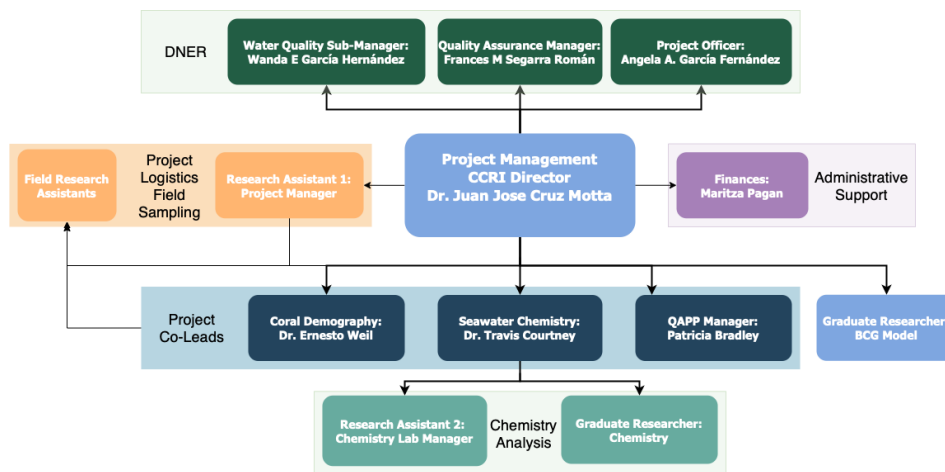


Figure 1: Organizational chart of the personnel involved in data collection, analysis and assessment.

Table 1: Program Roles and Responsibilities related to monitoring and data used with personnel contact information.

Personnel/Title and/or primary role	Responsibilities	Email Address	Phone
Dr. Juan J. Cruz Motta (CCRI director)	Provide authority and oversight of project, general management and personnel supervision, report editing and data analyses.	juan.cruz13@upr.edu	787-236-5376
Dr. Ernesto Weil (UPR DMS Director)	Training the students and personnel dealing with the coral demographic work. Responsible of coordinating the field work related to the demographic component the collection, organization and editing of the quantitative data.	ernesto.weil@upr.edu	787-375-5859

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Personnel/Title and/or primary role	Responsibilities	Email Address	Phone
Dr. Travis Courtney (CCRI researcher; Chemist)	Coordinating the water quality component of the project, overseeing data upload, helping to write reports, managing the chemistry lab, and supervising one of the research assistants	travis.courtney@upr.edu	910-512-7602
Mrs. Patricia Bradley (CCRI Board member; subcontractor)	Leading the QAPP design and coordinating the BCG work	patbradley@comcast.net	443-326-4884
Mrs. Maritza Pagan	Assisting with the financial aspects of the project	maritza.pagan@upr.edu	787-265-5408
Mr. Manuel Nieves, M.S. Research Assistant 1—(Project Manager M.Sc. level)	Overseeing the field personnel, logistics, data collection handling, editing and management, quality control and assurance. Assist with field work. Oversee the chain custody of samples. In close collaboration with CCRI director, will serve as liaison with DRNA and any other stakeholder. Serves as Dive Master.	manuel.nieves1@upr.edu To be hired	TBD
Dr. Martha Ricaurte Research Assistant 2 (CCRI Chemistry Lab Manager/Chemist M.Sc. level)	Manage the chemistry laboratory, all laboratory analyses, and assist with chemical measurements in the field as needed.	martha.ricaurte@upr.edu To be hired	TBD
Student assistantship	Assisting with field work and laboratory analyses	To be hired Various	Various TBD
Field Research Assistants (2)	Working diver. Provide expertise and technical support for water quality reef data collection. Assist in all diving operations.	Various To be hired	Various TBD
Wanda E. García Hernández, DNER's Water Quality Area Sub-Manager	Provide assistance in the evaluation of progress reports, assessment of compliance with Project objectives, and compliance with EPA's administrative requirements.	wandagarcia@drna.pr.gov	
Frances M. Segarra Román; DNER's Quality Assurance Manager (QAM)	In charge of Validating compliance with QA/QC protocols	francessegarra@drna.pr.gov	

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Monitoring Project in Shallow Coral Reef Areas around Puerto Rico**

Personnel/Title and/or primary role	Responsibilities	Email Address	Phone
Angela A. García Fernández; DNER's Project Officer	Liaison between DNER and CCRI personnel. In addition, in charge of evaluating progress report and assessing compliance with QA/QC requirements.	angelagarcia@drna.pr.gov	

Principal customers for the results of this research will be the following:

- DNER
- EPA Region 2
- Other State, Territory and Commonwealth Agencies that manage coral reefs
- NOAA
- Scientific community
- Environmental managers.

A3. PROJECT GOALS AND OBJECTIVES

The U.S. Clean Water Act (CWA) requires states and territories to regularly monitor, assess and report on the condition of their aquatic resources. States and territories are tasked under the CWA to adopt water quality standards that include criteria for acceptable physical, chemical, and biological conditions.

Puerto Rico DNER has the responsibility to implement public policy concerning the protection and conservation of the environment, using the resources necessary to prevent and eliminate damage that might affect it, wisely and judiciously. In 1999 the DNER established the Puerto Rico Coral Reef Monitoring Program (PRCRMP) with the following objectives

- Know the conditions of coral reef species with ecological and economic importance.
- Identify trends in reef communities in response to environmental and human pressures.
- Determine the most effective management strategies for reef protection.

In Puerto Rico, assessments of changes in water quality parameters in coral reefs have been limited to certain geographical areas and specific and very limited water quality parameters. In this regard, the identification of parameters that help sustain healthy coral reefs in Puerto Rico, is a pressing need due to the variability of environmental conditions and anthropogenic impacts that threaten the integrity of these marine ecosystems. DNER is interested in gathering water quality information in the sites included in the PRCRMP for shallow coral reefs areas (coral reef areas located at depths less than or equal to 30 meters) around Puerto Rico, quarterly, for two consecutive years.

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A4. PROJECT SCHEDULING & COORDINATION

DNER has contracted with UPR/DMS/CCRI to:

1. Design, plan, and execute a water quality monitoring program in the PRCRMP shallow coral reef sites. The monitoring sampling events will be performed quarterly at each monitoring station for two consecutive years. One of the sampling events shall be conducted within the period between one week before and one week after the coral health assessment of the PRCRMP is performed at each monitoring station.
2. Collect water quality information at the PRCRMP reef sites that are specified in the RFP (see Table 2)
3. Upload water quality data into the Water Quality Data Portal/WQX
4. Describe patterns of spatial and temporal variation of water quality parameters around Puerto Rico based on monitoring of the PRCRMP reef sites that are specified in the RFP (see Table 2)
5. Collect data on coral health and demographics at the PRCRMP reef sites in order to apply the BCG model
6. Assess covariation between the water quality parameters and results of the BCG model

A project schedule is shown below **(Figure 2)**.

Quarter	Q0			Q1			Q2			Q2			Q3			Q3			Q4			Q4			Q5			Q5		
Month	Initial	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26			
1. Project Planning																														
2. Develop QAPP																														
3. Establish and Verify SOP																														
4. Field Sampling																														
5. Sample Analysis																														
6. Upload Data																														
7. Reports																											F			

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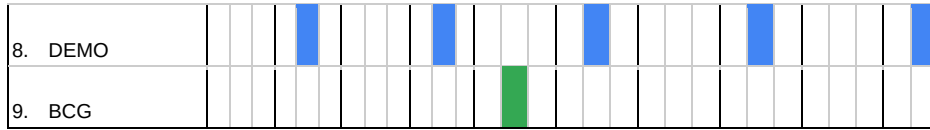


Figure 2: Project schedule and anticipated timelines following the initial QAPP approval and planning stages.

WQ sampling will be performed quarterly from small, shore-launched boats. Coral Demographics (DEMO) will be performed once at each site. Each PRCRMP site has a unique ID which identifies it throughout the pre-field, field, lab, analysis, and data management phases of the project. CCRI will procure, as needed, scientific collecting permits from Puerto Rico and Federal agencies. The field teams will use standard field equipment and supplies as identified in Appendix A. The Research Assistant 1 will coordinate equipment and supply requirements.

Water Quality.

The 42 PRCRMP reef sites listed in **Table 2** and shown in **Figure 3**, will be visited 4 times a year (approximately every three months) over a two year period for a total of 8 sampling times of 16 water quality parameters resulting in 64 data points per site per year and a total of 2,688 data points per year. Sampling effort has been allocated based on the region (**Table 2**) and it has been estimated that the following field days will be needed per region: Desecheo = 1 day, Culebra/Vieques = 3 days, North/East = 4 days, South and South/East = 3 days, South/West = 2 days, West = 7 days; for a total of 20 days per sampling cycle. Allocation of that sampling effort will be done haphazardly at the region level, to avoid confounding small scale variation with regional differences. At each sampling station, water quality will either be measured directly in the field or sampled and collected and analyzed at the CCRI Water Quality Monitoring Lab following the QAPP and best practices with respect to replicates, required analytical precision, and standards as outlined by their respective methods (**Table 3**). Bottom samples will be collected via direct measurement or Van Dorn water sampler.

Table 2: List of Puerto Rico PCRMP sites at which water quality and coral demographics data will be collected.

Site Name	Site Code	Location	Region	Zone	Central Latitude	Central Longitude
Boya Vieja (2015) 20m	15BOYA20	La Parguera	Southwest	SW	17.88827	-66.9978
Cabezas de San Juan 5m	18CABE05	Fajardo	North/Northeast	E	18.38544	-65.62972
Canal Luis Peña 5m	16LPEN05	Isla Culebra	Vieques/Culebra	E	18.30493	-65.32772
Canjilones (Vieques) 15m	01CANJ15	Isla Vieques	Vieques/Culebra	E	18.08964	-65.59008
Carlos Rosario 10m	16CROS10	Isla Culebra	Vieques/Culebra	E	18.32779	-65.332

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Site Name	Site Code	Location	Region	Zone	Central Latitude	Central Longitude
Cayo Aurora 5m	11AURO05	Guánica	Southwest	SW	17.9369	-66.8738
Cayo Caribes (2016) 10m	16CARI10	Salinas	Southeast	S	17.91533	-66.2141
Cayo Coral 10m	99CORA10	Guánica	Southwest	SW	17.9362	-66.8884
Cayo Diablo (2016) 5m	16DIAB05	Fajardo	North/Northeast	E	18.36033	-65.53089
Cayo Ratones 5m	16RATO05	Salinas	Southeast	S	17.9346	-66.3025
Cayo Rodriguez 5m	16RODR05	Mayagüez	West	W	18.1893	-67.1919
Cibuco (2013) 10m	13CIBU10	Vega Baja	North/Northeast	N	18.48955	-66.3736
Dakiti 20m	16DAKI20	Isla Culebra	Vieques/Culebra	E	18.27587	-65.2773
Derrumbadero 20m	01DERR20	Ponce	South	S	17.904	-66.6086
Dominos 5m	18DOMI05	San Juan	North/Northeast	N	18.46222	-66.0517
Efras Wall 20m	13EWAL20	La Parguera	Southwest	SW	17.8976	-66.9599
El Negro 10m	16NEGR10	Cabo Rojo	West	W	18.14653	-67.24803
El Negro 5m	16NEGR05	Cabo Rojo	West	W	18.14658	-67.24758
El Seco 30m	11SECO30	Isla Vieques	Vieques/Culebra	E	18.13869	-65.19714
Esperanza 10m	01ESPE10	Isla Vieques	Vieques/Culebra	E	18.08057	-65.48794
Guanajibo 20m	16GUAN20	Mayagüez	West	W	18.17202	-67.25297
Manchas Exteriores 10m	16MEXT10	Mayagüez	West	W	18.23353	-67.20057
Manchas Exteriores 20m	16MEXT20	Mayagüez	West	W	18.2335	-67.20092
Maria Langa 10m (2016)	16MLAN10	Guayanilla	South	S	17.96093	-66.75292
Maria Langa 20m (2016)	16MLAN20	Guayanilla	South	S	17.95953	-66.74698
Maria Langa 5m	16MLAN05	Guayanilla	South	S	17.96488	-66.75647

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Site Name	Site Code	Location	Region	Zone	Central Latitude	Central Longitude
Media Luna 10m (Parguera)	15MLUN10	La Parguera	Southwest	SW	17.93472	-67.0485
Media Luna 5m (Parguera)	15MLUN05	La Parguera	Southwest	SW	17.93937	-67.05067
Palominos (2016) 10m	16PALT10	Fajardo	North/Northeast	E	18.33537	-65.56555
Palominos (2016) 20m	16PALN20	Fajardo	North/Northeast	E	18.35466	-65.56711
Puerto Botes 15m	04BOTE15	Isla Desecheo	Mona/Desecheo	W	18.382	-67.48833
Puerto Botes 20m	00BOTE20	Isla Desecheo	Mona/Desecheo	W	18.38158	-67.4886
Puerto Canoas 30m	04CANO30	Isla Desecheo	Mona/Desecheo	W	18.37747	-67.484
Resuellos 10m	00RESU10	Cabo Rojo	West	W	17.99116	-67.23322
Tourmaline 10m	99TOUR10	Cabo Rojo	West	W	18.16304	-67.27367
Tourmaline 20m	04TOUR20	Cabo Rojo	West	W	18.16517	-67.27524
Tourmaline 30m	04TOUR30	Cabo Rojo	West	W	18.16637	-67.27632
Tres Palmas 10m	04TRES10	Rincón	West	W	18.34732	-67.26997
Tres Palmas 20m	04TRES20	Rincón	West	W	18.34652	-67.27118
Tres Palmas 30m	04TRES30	Rincón	West	W	18.35072	-67.26688
West Reef 10m	99CDMU10	Ponce	South	S	17.89502	-66.52838
Gallardo (2019) 5m	19GALL05	Cabo Rojo	West	W	18.0017	-67.3299

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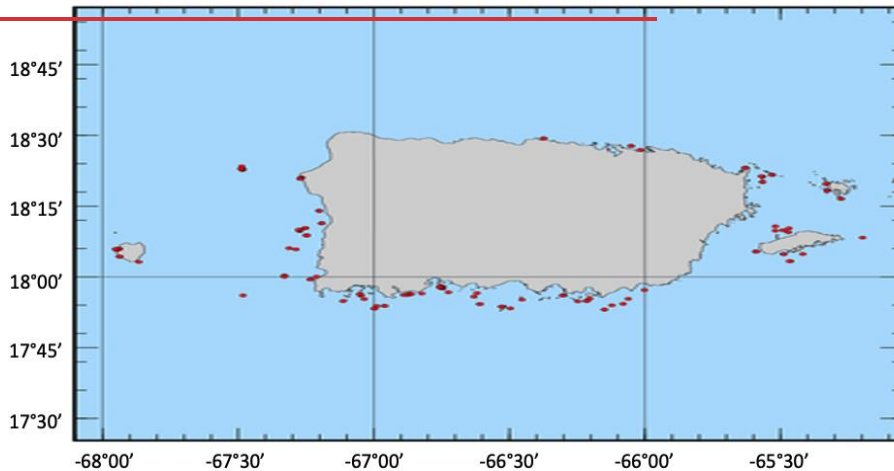
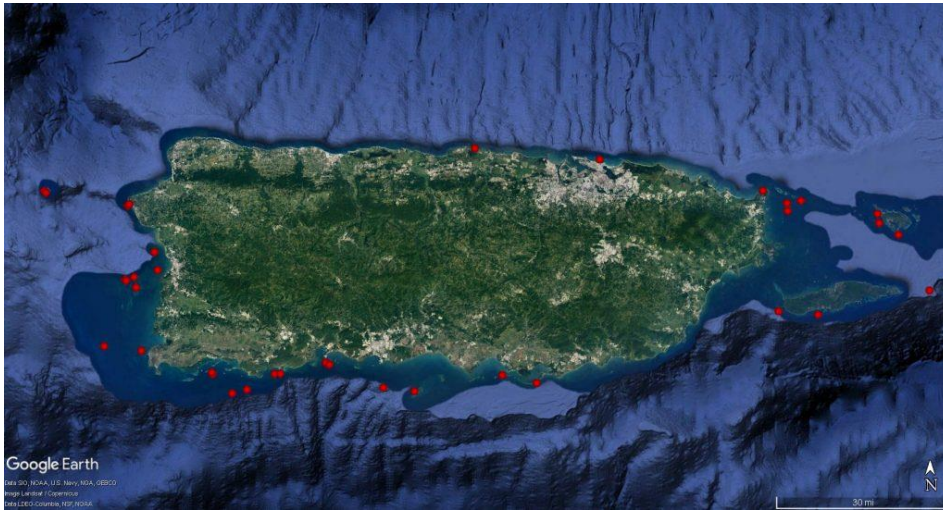


Figure 3: Map of Puerto Rico Coral Reef Monitoring Program (PRCRMP) sites. Each site is plotted as a red dot on the map of Puerto Rico. Adapted from NOAA National Centers for Environmental Information (<https://www.ncei.noaa.gov/access/metadata/landing-page/bin/iso?id=gov.noaa.node:0204647>).

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Table 3: Summary table of parameters, certified method (SM = Standard Method, EPA = United States Environmental Protection Agency), and location of measurement (i.e., in situ in the field, or sampled and analyzed in the Caribbean Coral Reef Institute Water Quality Monitoring Lab).

Parameter	Method	Field Protocol	Laboratory Protocol
Temperature	SM 2550 B	Surface and bottom measurements will be collected in the field (In-situ) using a Hydrolab HL4 Multiparameter Sonde (OTT HydroMet) <u>or comparable sonde (see discussion).</u>	N/A
BOD	SM 5210 B	Surface and bottom water samples will be collected into the field by completely filling the airtight sample container (BOD glass bottle with a Penny head stopper) from the water sampler sample container.	Determined using the difference in the DO measurements made before (initial DO) and after a 5-day incubation at 20°C and in the dark (final DO). Then, BOD is computed from the initial and final DO.
Chlorophyll-a	SM 10200 H	Bottom grab sample will be collected directly into the sample container; sample will be stored in amber glass bottles and transported to the laboratory inside a cooler with ice (6 °C).	Passed through a glass filter, which is then ground, dissolved in acetone, and analyzed on a AquaMate 8100 UV-VIS spectrophotometer)
DO	SM 4500-O H	Surface and bottom measurements will be collected in the field (In-situ) using a Hydrolab HL4 Multiparameter Sonde (OTT HydroMet) <u>or comparable sonde (see discussion).</u>	N/A
Enterococcus	SM 9230 B/C/D	Bottom sample will be collected directly into the sample container and transported to the laboratory inside a cooler with ice (6 °C).	The sample will be cultured and incubated for 24 hours to assess abundances as per the test method. Then, the colonies/100 mL will be determined by counting IDEXX Enterolert fluorescent wells in Quanti-Tray and Most Probable Number Table
pH	EPA 150.2	Surface and bottom measurements will be collected in the field (In-situ) using a Hydrolab HL4 Multiparameter Sonde (OTT HydroMet) <u>or comparable sonde (see discussion).</u>	N/A
Turbidity	EPA 180.1	Bottom sample will be collected directly into the sample container	Nephelometry on a calibrated Orion AQ4500 Portable Turbidimeter)

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Parameter	Method	Field Protocol	Laboratory Protocol
		and transported to the laboratory inside a cooler with ice.	
Salinity	SM 2520 B	Surface and bottom measurements will be collected in the field (In-situ) using a Hydrolab HL4 Multiparameter Sonde (OTT HydroMet) <u>or comparable sonde (see discussion).</u>	N/A
Secchi Depth	Secchi Disk	Distance, for water transparency, following standard procedures)	N/A
Settleable Solids	SM 2540-F	Bottom grab sample will be collected directly into the sample container and transported to the laboratory inside a cooler with ice (6 °C).	Deposited into an Imhoff Cone for the fixed 1-hour settling period to quantify the total amount of settleable solids
TSS	SM 2540-D	Bottom grab sample will be collected directly into the sample container and transported to the laboratory inside a cooler with ice (6 °C).	Filtering a fixed amount of seawater and determining the change in weight of the filter
TP	EPA 365.41	to improve sample preservation Bottom grab sample will collected directly into the sample container, preserved in the field with H₂SO₄ (pH < 2.0); and transported to the laboratory inside a cooler with ice (6 °C).	Quantified in the lab using a SEAL AutoAnalyzer AA500
TKN	EPA 351.21	All nutrient samples (TP, TKN, Nitrate+Nitrite) will be filtered directly in the field and acidified to improve sample preservation Bottom grab sample will collect directly into the sample container, preserved in the field with H₂SO₄ (pH < 2.0); and transported to the laboratory inside a cooler with ice (6 °C).	Quantified in the lab using a SEAL AutoAnalyzer AA500. Total Nitrogen will be determined as the sum of TKN and Nitrate + Nitrite.
Nitrate+Nitrite	EPA 353.2	Bottom grab sample will collect directly into the sample container and transported to the laboratory inside a cooler with ice (6 °C).All nutrient samples (TP, TKN, Nitrate+Nitrite) will be filtered directly in the field and acidified to improve sample preservation Bottom grab sample will be taken -- acidified in the field and kept cool	Quantified in the lab using a SEAL AutoAnalyzer AA500. Total Nitrogen will be determined as the sum of TKN and Nitrate + Nitrite.

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Parameter	Method	Field Protocol	Laboratory Protocol
Temperature and Salinity Depth Profiles		Surface and bottom measurements will be collected in the field (In-situ) using a Hydrolab HL4 Multiparameter Sonde (OTT HydroMet).	N/A
TA	Dickson et al., 2007	Grab samples collected <u>at</u> the surface and at specific depth, directly into the sample container, preserved in the field with HgCl ₂ , and transported to the laboratory inside a cooler with ice (6 °C).	Gran titration for total alkalinity will be performed using an Apollo AS-ALK3 following best practices <u>and manufacturer recommendations.</u>
DIC	Dickson et al., 2007	Grab samples collected at the surface and at specific depth, directly into the sample container, preserved in the field with HgCl ₂ , and transported to the laboratory inside a cooler with ice (6 °C).	Infrared detection of CO ₂ liberated by acid will be performed using an Apollo AS-C5 Model DIC Analyzer, following best practices and <u>manufacturer recommendations.</u>

Deliverables:

1. Work Plan and Time Schedule
2. Quality Assurance Progress Plan (QAPP)
3. Maps in GIS Shapefile format and KML format of the monitoring stations of the monitoring effort, including the metadata.
4. Quarterly invoices for the activities performed during the period, will be certified by the researcher in charge and they will detail the tasks corresponding to the respective period. These invoices will accompany each one of the progress reports. Original receipts will be attached to these invoices. The progress reports must include the field documentation, as well as the Certificate of Analysis of the laboratory test performed to each one of the samples collected, in an electronic or digital format.
5. Progress seminars will be delivered by CCRI as needed to inform stakeholders of progress and main results of this project. This activity was not requested in the RFP, but will be offered by CCRI as in kind, as it constitutes one of its main missions (i.e., outreach and education).

A5. DATA QUALITY OBJECTIVES AND CRITERIA

The U.S. EPA requires that Data Quality Objectives (DQOs) be developed for all environmental data collection activities following the prescribed DQO Process. DQOs are qualitative and quantitative statements that clarify study objectives, define the appropriate types of data, and specify the tolerable levels of potential decision errors that will be used as the basis for establishing the quality and quantity of data needed to support decisions (EPA 2006B). Data quality objectives thus provide the criteria to design a sampling program within cost and resource constraints or

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technology limitations imposed upon a project or study. DQOs are typically expressed in terms of acceptable uncertainty (e.g., width of an uncertainty band or interval) associated with a point estimate at a desired level of statistical confidence (EPA 2006B).

The DQO Process is used to establish performance or acceptance criteria, which serve as the basis for designing a plan for collecting data of sufficient quality and quantity to support the goals of a study (EPA 2006B). As a general rule, performance criteria represent the full set of specifications that are needed to design a data or information collection effort such that, when implemented, generate newly-collected data that are of sufficient quality and quantity to address the project's goals (EPA 2006B). Acceptance criteria are specifications intended to evaluate the adequacy of one or more existing sources of information or data as being acceptable to support the project's intended use (EPA 2006B).

Data Quality Objectives for the CCRI Water Quality Monitoring Project in Shallow Coral Reef Areas around Puerto Rico:

The procedures established in this QAPP are to ensure that the samples results are:

1. Representative of the location sampled.
2. Reliable, scientifically valid, of known precision and accuracy, and of acceptable completeness and representativeness, in order to generate data of the highest quality.
3. Reproducible.

Measurement Quality Objectives

The first objective will be met by the performance of consistent field activities in compliance with all the procedures described in this QAPP and by examining data collected during these 8 surveys to measure water column parameters, quantify nutrients and chlorophyll, and document changes in stony coral community structure and condition. This includes using the best sampling practices to ensure a representative sample. In the case of the second objective, it will be met by ensuring that measurements are recorded, and all samples are sampled continuously in compliance with the procedures described in this QAPP, and making sure all samples and measurements are collected at the same locations over time to quantify the variability of results at each station. Finally, the third objective will be met by analyzing laboratory replicates to ensure reproducibility of results. Definitions of quality control samples are provided in Section B5 of this document.

Quality Requirements of Water Sampling and Analysis

Precision and Accuracy

Precision and accuracy of laboratory procedures are ensured by the analysis of quality control (QC) samples, including method/filter blanks, prepared standards, standard reference samples (SRS); and whenever available or applicable, laboratory control samples, laboratory replicates, matrix spike/matrix spike duplicate (MS/MSD), field duplicates, and field replicates, will be used also to assess them. **Table 4** lists the desired precision, accuracy, and detection limit goals for each

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parameter to be measured. QC samples and method procedural blanks to be analyzed in the laboratory to assess precision and accuracy are listed in the respective EPA methods (**Table 4**) and will be further detailed in the respective SOP for each parameter.

For field activities, accuracy (a measure of confidence in a measurement) will be evaluated through the validation of the field instruments' calibration verification process (calibration verification form). For analytical laboratory procedures accuracy will be assessed through calibration and calibration verification process, laboratory blanks, and quality control samples (spikes), as applicable. Percent Recovery (%R) for all spikes samples will be calculated as follows:

$$\% R = \frac{\text{Result of the control sample}}{\text{Spike analyte concentration added to fortify}} \times 100$$

If %R is outside of the acceptance criteria of the test method, the appropriate corrective action should be undertaken.

For the bacteriological parameter, quality control will be conducted on each new lot of Enterolert, sample containers and Quanti-trays. To assure the quality of the analysis three sterile vessels filled with 100ml sterile water will be inoculated with *Enterococcus faecium* ATCC 35667 (fluorescence), *Serratia marcescens* (gram negative) ATCC 43862 (no fluorescence) and *Aerococcus viridans* (gram positive) ATCC 10400 (no fluorescence). Enterolert media is added to each vessel that are poured in a Quanti – tray for incubation (41°C for 24 hours). In addition, blank samples of sterile dilution water are included. These controls will be received from the manufacturer. Finally, at least one external reference sample (Proficiency Test) must be run per year.

Precision is the measurement of agreement among repeated measurements. For sampling activities, precision will be assessed using the duplicate samples, which will be collected in a frequency of 1 every ten (10) samples. For the analytical test that established it in the specific test method, the precision will be evaluated by analyzing Laboratory Blank Spike Duplicate. Precision for the field duplicates and laboratory spike duplicates will be assessed by calculating RPD.

$$RPD = \frac{\text{Original sample result} - \text{Duplicate sample result}}{\left(\frac{\text{Original sample result} + \text{Duplicate sample result}}{2} \right)} \times 100$$

If RPD is greater than the acceptance criteria of the test method, appropriate corrective action should be undertaken.

Comparability

Data will be directly comparable to results obtained from similar sites in Puerto Rico because field program design and analytical procedures are similar or identical. In addition, use of written standardized procedures and EPA approved methods ensures that sample preparation and analyses will be comparable throughout the project and with other projects following EPA methods. As part of the data assessment, field and analytical results will be evaluated to determining if:

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1. Sample collection and handling methods were performed in accordance with requirements and procedures established in this QAPP and the specific test method.
2. Field equipment calibration was carried out following the procedures established in this QAPP, the specific test method or SOP, and by the manufacturer.
3. Holding times were followed as recommended by the analytical methods references and established in Table 5 of this QAPP.
4. Analytical procedures were followed as established in the methods references and laboratory SOPs.
5. In addition, comparability will be ensured by using measurement units consistently throughout the reports.

Representativeness

Representativeness is the degree to which accurate and precise data represents the characteristics of a statistical population, the parameter variations at a sampling point, a process condition, or an environmental condition. Representativeness is addressed in the PRCRMP sampling design. The sampling practices and laboratory measurements that will be performed during the water quality monitoring have already been used in many coastal and marine systems to characterize water quality and coral demographics and are, therefore, expected to yield data representative of the study area. Representativeness will also be ensured by proper handling, storage (including appropriate preservation and holding times), and analysis of samples so that the material analyzed reflects the material collected as accurately as possible. The following will be considered or performed to assess and ensure the representativeness of the data:

1. Whether there are enough sampling locations, matrices, and samples to accurately depict site conditions.
2. Establishing sampling procedures designed to collect individual samples that accurately represent the chemistry of the matrix from which they were collected.
3. Whether or not the appropriate analytical test method was used based upon the type of sample collected and the data quality objective established in this QAPP.
4. Data Quality Assessment and Verifications (DQAV) to review the adherence to field and laboratory procedures to assess the representativeness of the field and analytical results.

If the data generated do not meet the quality objectives of this project, it will be flagged and reviewed to determine if appropriate quality controls are in place. Deviations from the analytical scheme described in this QAPP will be noted in the laboratory records associated with analytical batches in the QA statements, and the appropriated corrective actions will be implemented.

Sensitivity

Sensitivity is the capability of methodology or instrumentation to discriminate among measurement responses for quantitative differences of a parameter of interest. In addition, it is the

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ability of the method or instrument to detect the target analytes at the level of interest. The method detection limits (MDLs) provide the sensitivity goals for the procedures as outlined in **Table 4**.

Data users should be aware that precision and accuracy generally degrade as analyte concentrations decrease. While numerical results are being reported down to the MDL, results below the lowest calibration standard will often have precision and accuracy that don't meet the data quality objectives for the project.

Completeness

It is expected that 100% of samples collected for analysis will in fact be analyzed. However, a sample loss of < 10% will not compromise the objectives of the project.

Table 4: Desired precision, accuracy, and MDL for each parameter defined by the limits of detection for the respective method, instrumentation, and certified standards based on quality objectives.

Parameter	Method	Units	MDL	Accuracy	Precision
Temperature	SM 2550 B	°C	-5.0°C	± 0.1°C	<20%
				± 20%	
BOD	SM 5210 B	mg/L	0 mg/L	± 0.2 mg/L	<20%
				± 20%	
Chlorophyll-a	SM 10200 H	µg/L	5 µg/L	N/A	<20%
DO	SM 4500-O H	mg/L	0.01 mg/L	± 0.2 mg/L	<20%
				± 20%	
Enterococcus	SM 9230 D	ind/100mL	1/100mL	1/100mL N/A	<60%
pH	EPA 150.2	N/A	-2	± 0.00102	<10%
				± 10%	
Turbidity	EPA 180.1	NTU	0.01 NTU	± 20% ± 0.01 NTU	<20%

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Parameter	Method	Units	MDL	Accuracy	Precision
Salinity	SM 2520 B	psu	0.06 psu	± 0.50 psu	<20%
				± 20%	
Secchi Depth	Secchi Disk	m	N/A	± 20%	<20%
Settleable Solids	SM 2540-F	mL/L	0.1 ml/L	N/A	<20%
				±20%	
TSS	SM 2540-D	mg/L	2.5 mg/L	± 5 mg/L	<20%
				± 20%	
TP	EPA 365. <u>14</u>	mg/L	0.0 503 mg/L	± 0.0 503 ⁹⁴ mg/L	<20%
				± 20%	
TKN	EPA 351. <u>24</u>	mg/L	0.0 503 mg/L	± 0.0 503 ⁹⁴ mg/L	<20%
				± 20%	
Nitrate + Nitrite	EPA 353.2	mg/L	0.0 503 mg/L	± 0.0 503 ⁹⁴ mg/L	<20%
				± 20%	
TA	Dickson et al. (2007)	μmol/kg	N/A	± 3μmol/kg	<20%
				± 20%	
DIC	Dickson et al. (2007)	μmol/kg	N/A	± 3μmol/kg	<20%
				± 20%	

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Special Training Requirements and Certification

WQ Monitoring.

WQ sample collection requires no non-routine field sampling techniques, field analyses, laboratory analyses, or data validation. Field personnel are experienced in using the equipment identified and/or will be trained by the supervising chemist prior to any field sampling and lab analyses.

Coral Demographics Data Collection.

Coral demographics surveyors require both specialized training and certifications. Detailed training and certification is provided in Section A8.

Laboratory Analyses.

All chemistry measurements use routine laboratory analyses as documented in the respective SOP for each parameter. Lab personnel will be trained in standard protocols specified in PCCS's Laboratory Quality Assurance Plan for handling, storing, and preparing samples for analysis. Laboratory personnel will be trained in using the equipment identified within this QAPP following SOP. The supervising Chemist is a Puerto Rico certified Chemist, which will oversee the analysis of seawater in this project and certified the analytical test results as required by Law and Regulations.

Documentation and Records

CCRI will maintain all documents relevant to sampling, laboratory analysis, and data analysis activities. All data will be archived electronically and backed up online. Hard copies of field and lab notebooks will be archived in CCRI facilities at UPRM on Isla Magueyes. Hard copies of data will be kept for at least one year following the termination of the contract.

Document Control

CCRI will maintain documents relevant to laboratory analysis activities and data entry. Document Control is the responsibility of the Program Manager. A copy of the most current analyses SOP is kept in the lab area where the analysis is being performed. This document references the SOP number without the revision number. All members of the project team will inform the CCRI Program Manager of the need for SOP revisions. All the documents sent and kept at the DNER, will be controlled, managed and stored as per the DNER internal document retention procedures and policies. Electronic copies of the SOP will be provided to the DNER.

Laboratory Analyses Records

All data will be recorded initially into bound laboratory logbooks, onto established data forms (Appendix F) or onto electronic file where applicable. All the analytical laboratory test results sent to the DNER will be kept in the files as established by the department's internal document retention procedures and policies. Electronic copies of the Certificate of Analysis will be provided to the DNER in the progress reports.

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Records Retention and Storage

All data will be archived electronically and backed up online. Recent hard copies of field and lab notebooks will be kept in the designated file cabinet at CCRI. Archived records will be stored in the data vault located in CCRI facilities at UPRM on Isla Magueyes for at least one year following the termination of the contract. All the documents sent to the DNER will be kept in the files as established by the department's internal document retention procedures and policies.

A6. TRAINING AND CERTIFICATION

A successful DEMO survey depends on the abilities of the survey team to locate the PRCRMP fixed survey, establish the radial belt transect, conduct surveys, document results on underwater slates, recover underwater equipment and return safely to the boat. To accomplish these activities, all surveyors will be AAUS registered divers, possessing a certificate of first aid training in Cardiopulmonary Resuscitation, and a certificate in oxygen administration. All divers will have had a physical examination and current tetanus inoculation. Eligibility of all divers will be reviewed by the Diving Officer (DOD) prior to embarkation and status may be reviewed by the DOD and Chief Scientist at any time during the project.

Surveyors must be able to identify corals to genus and species, measure coral colony dimensions and estimate the proportion of tissue in relation to the overall size of the colony. Surveyors must be able to identify bleaching, disease, invasive growth of clionid sponges, and predation damage. (For detailed disease, predation, bleaching and overgrowth descriptions see regional specific websites: GCDD 2012; NOAA CDHC 2012; NOAA CORIS 2012). All members of each field team will be trained by CCRI before the field season. Field sampling audits or evaluation visits will be completed for each field team. The training program will stress hands-on practice of methods, consistency among crews, collection of high-quality data and samples, and safety.

A7. DOCUMENTATION AND RECORDS

Data will be collected on standard forms developed for the studies and printed on Duracopy™ Waterproof Paper. Upon return to shore, surveyors and photographers will photograph and rinse their underwater data sheets in fresh water and hang them to dry in a dedicated, safe location. Coral survey data will be entered into a standardized electronic data sheet developed in MSExcel®. Records will be entered by someone other than the surveyor and verified for accuracy by the surveyor. The database will be compiled and backed up on the cloud each evening in a separate file. The project manager will be responsible for record keeping and documentation of procedures and data in accordance with the CCRI Records Management Plan. Survey data sheets will be photographed and retained by the quality assurance manager. Digital images of corals will be copied on hard drives and retained by the quality assurance manager.

Lab data will be retained in laboratory notebooks, the respective instrument's data logs, and online data repositories. See subsequent data management section.

SECTION B: DATA GENERATION AND ACQUISITION

B1. SAMPLING PROCESS DESIGN

The PRCRMP reef sites listed in **Table 2** will be visited every three months during a two-year period, for a total of eight (8) WQ sampling events. Sampling effort has been allocated based on the region (Table 1) and it has been estimated that the following field days will be needed per region: Desecheo = 1 day, Culebra/Vieques = 3 days, North/East = 4 days, South and South East = 3 days, South West = 2 days, West = 7 days; for a total of 20 days per sampling cycle. Allocation of that sampling effort will be done randomly at the region level, to avoid confounding small scale variation with regional differences.

DEMO surveys will be performed once at each PRCRMP reef site, along each permanent transect (3 transects at each site).

B2. SAMPLING METHODS

Field measurements and samples will be collected by trained survey teams and in accordance with the procedures listed in Appendix J and will be the basis for the field SOP that will be submitted in electronic format to DRNA, before starting field operation. For each site, crews prepare a dossier that contains the following applicable information: scientific collection permits, coordinates of the coastal site, and local area emergency numbers. Whenever possible, field team leaders attempt to contact owners of private boats and/or marinas or boat launches (as appropriate) approximately two days before the planned sampling date. As the design requires repeat visits to select sampling locations, it is important for the field teams to do everything possible to maintain good relationships with boat/launch owners. This includes prior contacts, respect of special requests, closing gates, minimal site disturbance, and removal of all materials, including trash, associated with the sampling visit.

Field Measurements

Standardized field data forms (Appendix F) are the primary means of data recording. On completion, the data forms are reviewed by a person other than the person who initially entered the information. Prior to departure from the field site, the field team leader reviews all forms and labels for completeness and legibility and ensures that all samples are properly labeled and packed. Forms are photographed following each day and uploaded to CCRI servers to ensure a digitized version remains available in addition to the paper form.

Upon return to the office from field sampling events, completed data forms are sent to the CCRI Project manager (or delegated assistant) for entry into a computerized database. These forms must be sent within 2 weeks from the sampling event. The Project manager will ensure that electronic data files are reviewed independently to verify that values are consistent with those recorded on the field data form or original field data file.

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The CCRI Water Quality Monitoring Lab will adhere to the provisions of this integrated QAPP and will follow best laboratory practices. The CCRI Water Quality Monitoring Lab Data Management Plan is provided in Appendix G. For each parameter the proponent will use an analytical method approved by EPA with a detection limit below the applicable water quality standard (**Table 3**; Reference: Puerto Rico Water Quality Standard Regulation, 2019).

Water Quality Parameters and Sampling

Once in the field, the boat captain navigates to the site using a handheld GPS unit. At each sampling site, water quality parameters will either be measured (in-situ measurements) or a water sample will be collected and analyzed at the Caribbean Coral Reef Institute Water Quality Monitoring Lab. In the case of water samples, they will be collected as per the sampling method described in this QAPP and in the frequency and amount established for replicates, required analytical precision, and standards as outlined by their respective methods (**Table 3**). Surface and subsurface samples will be collected via a Van Dorn water sampler unless otherwise required by the stated methods (**Table 3**). See Appendix J for further details.

Samples are stored or packaged for shipment in accordance with instructions contained in the Field Operations Manual (EPA, 2010A). Precautions must be taken to ensure that the holding times are not exceeded for the samples that must be delivered to the CCRI Water Quality Monitoring Lab. The Project Coordinator will be notified to track the samples; thus, tracing procedures can be initiated, and corrective actions can be implemented as soon as possible, in the event samples are not received. Chain-of-custody forms must be properly completed for all transfers of samples, with copies maintained by the field team. In addition, any entry error must be corrected by crossing the text with a line and writing the corrections with the date and initials of the person making the corrections.

The CCRI Water Quality Monitoring Lab will adhere to the provisions of this integrated QAPP and will follow best laboratory practices. The CCRI Water Quality Monitoring Lab Data Management Plan is provided in Appendix G. For each parameter the proponent/contractor will use an analytical method approved by EPA with a detection limit below the applicable water quality standard (**Table 4**; Reference: Puerto Rico Water Quality Standard Regulation, 2019). The CCRI Water Quality Monitoring Lab is currently procuring equipment and developing SOP for each parameter investigated in this study to ensure accuracy and precision are met (**Table 4**) following approved methods (**Table 3**) and will be finalized, tested, and verified as part of the first stage of the QAPP to ensure all standards are met for this monitoring project (**Table 4**; Reference: Puerto Rico Water Quality Standard Regulation, 2019). Final SOPs will be submitted electronically to DNER as soon finished using the official CCRI format.

Samples will be processed and analyzed as designated batches consisting of 25 or less samples and each batch will include prescribed QC samples (e.g., reagent blanks, matrix spikes and matrix spike duplicates, and standard reference materials (SRMs)). These QC samples represent the basic elements that provide estimates of accuracy and precision for the analyses of chemical contaminants. The overall analytical process involves several additional QC- related components or checks (e.g., calibration curves, standards, reagent blanks, duplicates, and standard reference materials) that will be run with each batch of samples. If the prescribed quality criteria are not

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consistently met, the analyst will confer with the laboratory supervisor for corrective measures before proceeding with additional samples.

When these QC checks are embedded in each batch, the analyst should be able to quickly assess the overall data quality on a per batch basis and take corrective measures if there are deficiencies. If data for a class of compounds consistently fails any of the designated quality standards, the laboratory management must notify the CCRI Project Manager of the problem and seek recommended corrective actions prior to submitting the final data report.

The CCRI Water Quality Monitoring Lab will calculate and document method detection limits (MDLs) for each analyte of interest for each matrix which they plan to analyze, following the procedure specified in 40 CFR Part 136 (Federal Register, Oct. 28, 1984).

Coral Demographics.

This information will complement that collected by PRCRMP. PRCRMP started in 1999 with baseline characterizations of reef substrate cover by sessile-benthic categories and determinations of fish and motile megabenthic invertebrate taxonomic composition and densities. Annual monitoring events were conducted on a total of 15 reefs between 2003 and 2015. In 2016, the PRCRMP was modified to expand the geographical range and number of reef monitoring sites to achieve a 42- reef station framework, with alternate year monitoring surveys at each reef station. PRCRMP determines the percent substrate cover by reef substrate (benthic) categories from sets of five replicate 10m-long permanent transects per reef station based on a continuous (chain-link) intercept technique.

CCRI will conduct DEMO surveys using the same five transects at each PRCRMP station, following a similar schedule used by PRCRMP, that is all sites will be assessed only once during the duration of this project. PRCRMP assesses their sites every other year, consequently half of their sites are done each year.

The DEMO method is used to observe and record coral reef community condition. Diver's swim along the permanent, tagged transects laid over hard-bottom and coral reef habitats selected. The DEMO information will be collected using a different method however, instead of a chain-transect, a 10m long x 1m wide band transect will be used, and all live coral colonies larger than 5cm in diameter in the band transect will be measured (diameters and height). The information to be collected includes species composition, live coral cover, species density and size structure, juvenile abundance, and specific parameters of health condition (% live vs. dead, disease, and bleaching) of the overall coral community (all corals species in the band-transect). From the species, size, and mortality measures of the DEMO surveys, coral surface area (CSA), and live coral surface area (LCSA) can be calculated in 2 and 3 dimensions.

Detailed methodology (based upon the current NOAA National Coral Reef Monitoring Program (NCRMP) coral demographics protocol) is provided in **Appendix A**. The NOAA National Coral Reef Monitoring Program (NCRMP) coral demographics protocol provides more detailed and species-specific insight for coral populations than is provided by percent cover (as currently

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monitored by PRCRMP). Specifics of the protocol are based closely on other long-established monitoring programs in the Atlantic region, including Atlantic and Gulf Rapid Reef Assessment (AGRRA, Caribbean-wide), EPA's Stony Coral Rapid Bioassessment Protocol (Fisher 2007); Sanctuary Coral Reef Ecosystem Assessment and Monitoring (SCREAM, Florida), Florida Reef Resilience Program (FRRP, Florida), Coral Reef Evaluation and Monitoring Project (CREMP, Florida), and the U.S. Virgin Islands Territorial Coral Reef Monitoring Program (TCRMP, USVI). **Appendix B** provides a list of coral species and their CCRI codes. **Appendix C** is an example of the CCRI boat/diver log sample. **Appendix D** is the Coral Demographic Datasheet. **Appendix E** provides a list of habitat types.

B3. SAMPLE HANDLING AND CUSTODY

The purpose of the sample custody records is to document, maintain the integrity, and record the routing of all water quality samples during transportation and laboratory login operations. A record of all transactions concerning each sample will be maintained. The Field Sampling Leader is responsible for the care and custody of the samples until they are properly transferred to the CCRI Chemistry Laboratory for analysis. The QAPP Manager determines whether proper custody procedures were followed during field operation.

The Chain of Custody (COC) documents that will be completed during each monitoring event is presented in **Appendix H**. The COC will remain in the field samples coolers to document sample transfer. The CCRI Chemistry Laboratory will return a completed copy of the form after the samples are delivered.

A field data sheet will be completed on-site at the time of sampling (see **Appendix F**). All samples will be placed in appropriately labeled containers matching the data entry sheet for delivery to the CCRI Water Quality Lab. Sample volumes, sample preservatives, and holding times are presented in Table 5. Small sample containers will be sealed in plastic "zip-lock" bags in order to protect the labels from water damage and to ensure that the containers are not in direct contact with the ice blocks. Water samples will be placed in a sturdy ice cooler and chilled to approximately 4°C for transit to the CCRI Water Quality Lab. The samples will be logged by the laboratory upon receipt.

Table 5: Sample volumes, preservatives, and holding times for each parameter.

Parameter	Method	Sample Volume	Preservation	Holding Times
Temperature	SM 2550 B	In-situ	N/A	Immediately after collection
BOD	SM 5210 B	300 mL 1 L ^{ab}	6 °C	48 hours
Chlorophyll-a	SM 10200 H	1 L ^{be}	6 °C	24 hours
DO	SM 4500-O H	In-situ	N/A	Immediately after collection

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Parameter	Method	Sample Volume	Preservation	Holding Times
Enterococcus	SM 9230 ^D	250 mL ^a Whirl-Pak	6 °C	6 hours
pH	EPA 150.2	In-situ	N/A	Immediately after collection
Turbidity	EPA 180.1	250 mL ^a	N/A	48 hours
Salinity	SM 2520 B	In-situ	N/A	Immediately after collection
Secchi Depth	Secchi Disk	In-situ	N/A	Immediately after collection
Settleable Solids	SM 2540-F	1 L ^a	N/A	48 hours
TSS	SM 2540-D	1 L ^a	N/A	48 hours
TP	EPA 365. ¹⁴	500 15 125 mL ^{cd}	H ₂ SO ₄ , pH<2.0, 6 °C	28 days
TKN	EPA 351. ²⁴	500 mL ^{de}	H ₂ SO ₄ , pH<2.0, 6 °C	28 days
Nitrate + Nitrite	EPA 353.2	125 125 mL ^{cd}	H ₂ SO ₄ , pH<2.0, 6 °C	28 days
TA	Dickson et al (2007)	125 50 mL ^{fe}	HgCl ₂ , 6 °C	28 days
DIC	Dickson et al (2007)	125 50 mL ^{ef}	HgCl ₂ , 6 °C	28 days

Notes:

^a Polypropylene plastic bottle. ^b BOD Bottle (glass bottle with Penny head stoppers)

^{be} Amber glass bottle

^{cd} Single 125 mL polypropylene plastic bottle 15 mL falcon tube to minimize vessel contamination for sample analyses for both TP and Nitrate + Nitrite

^{de} Single 50 mL falcon tube to minimize vessel contamination for sample analyses

^{ef} Single 150 mL Pyrex Borosilicate screw-top glass bottle; One bottle can be taken and analyzed for both TA and DIC

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B4. ANALYTICAL METHODS

Seawater temperature, DO, salinity, and pH measurements will be collected using a Hydrolab HL4 following the test methods and the field and laboratory protocols indicated in Tables 3 and 5. In the event of instrument failure, an alternative sonde with comparable specifications, precision, and

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accuracy, may be used as defined in the SOP for T, S, DO, and pH. Secchi distance will be recorded by a trained observer with a Secchi disk at each station following standard procedures (Table 3, 4, 5; Appendix J). In addition to collecting and analyzing these required samples as part of the BCG model and Water Quality Board of Puerto Rico, we propose to collect temperature and salinity depth profiles via a Hydrolab HL4 to quantify any stratification at each of the sampling locations that may be influencing underlying biogeochemical parameters (Table 3).

During the sampling events, samples will be collected into the sample containers indicated in Table 5 and using the Field Protocols indicated in Table 3. These samples will be analyzed for the parameters indicated in Tables 5 (i. e. BOD, Chlorophyll-a, Enterococcus, Turbidity, Settleable Solids, TSS, TP, TN, Nitrate+Nitrite, TA and DIC) by the laboratory using the test methods and laboratory protocols indicated in Tables 3 and 5.

Additionally, samples will be collected for total alkalinity (TA) and dissolved inorganic carbon (DIC) at surface and depth and analyzed as indicated in Tables 3 and 5. Reef seawater TA and DIC will be assessed at depth relative to ~~offshore~~ surface TA and DIC ~~collected and analyzed following the same procedures~~ to evaluate relative coral reef net ecosystem calcification (NEC) and net ecosystem production (NEP) using a vector approach as discussed by Cyronak et al. (2018). In particular, this chemical evaluation of coral reef metabolism (i.e., NEC and NEP) through time provides additional information regarding the relative reef-building capacity of each location and the relative contributions of organic to inorganic carbon cycling, which can provide an indication of shifting reef processes as a result of benthic community changes (Cyronak et al. 2018).

B5. QUALITY CONTROL

All QA and QC procedures for the chemistry parameters and the desired accuracy and precision thresholds for blanks, duplicates, matrix spikes, lab control samples, and certified reference materials are defined in Table 5 and the subsequent EPA and SM methods.

Activities of the project include characterization of stony coral assemblages. Quality data is ensured by training of underwater surveyors (see SECTION A9) and quality control of within-sample variability will be examined through repetition at single sites. Data are entered by CCRI personnel other than the surveyor and verified by the surveyor within 24 h. Upon return, data in computer printouts will be compared to raw data sheets for at least 10% of the stations surveyed.

B6. FIELD INSTRUMENT/EQUIPMENT TESTING, INSPECTION AND MAINTENANCE

Instrument Inspection and Maintenance

The Hydrolab HL4 Multiparameter Sonde (OTT HydroMet) to be used for temperature, salinity, DO, and pH measurements will be calibrated and maintained by the CCRI Chemistry Laboratory and in accordance with the manufacturer recommendations, the corresponding laboratory's SOP, and the procedures describe below. All field equipment including the Secchi disk will be maintained according to the manufacturer instructions and will arrive each day in proper working

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conditions. After each monitoring event, the Hydrolab HL4 probes will be inspected for damage. Particular attention will be given to the dissolved oxygen probe membrane for the observation of wrinkling or other physical damage. This membrane will be changed according to the manufacturer's maintenance schedule. Sampling equipment will be checked for visual material and mechanical integrity on each day of use. When not in use the Hydrolab HL4 will be stored in the CCRI Chemistry Lab. The instruments will be secured, padded, and protected during transportation.

Instrument Calibration and Frequency

Temperature, salinity, DO, and pH measurements

The Hydrolab HL4 Multiparameter Sonde (OTT HydroMet) to be used for temperature, salinity, DO, and pH measurements ~~will be calibration will be checked~~ at the beginning of each sampling day following manufacturer recommendations and the corresponding field or laboratory SOP. In the event of instrument failure, another sonde may be used following the same calibration procedures with similar accuracy and precision as needed. Also, the calibration data will be recorded in the calibration sheets (Appendix I). The calibrations will be assessed relative to the glass filled thermometer, salinity standard, 100% saturation, and pH=7 standard at the ~~end-start~~ of each sampling day to ensure less than 20 % instrument drift ~~in one between~~ sampling days, ~~as Table 6:~~

Table 6: Hydrolab HL4 Multiparameter Sonde (OTT HydroMet Calibration Frequency, Conditions, Verification, and Acceptance Criteria.

Sonde parameter	Calibration Frequency	Calibration Solutions	Calibration Verification	Acceptance Criteria	Corrective Action
Temperature	Factory set	N/A	Will be assessed for accuracy relative to a certified glass thermometer.	$\pm 0.1^{\circ}\text{C}$	1. Call manufacturer 2. Send to factory for instrument servicing
Salinity	Daily	Dry sensor (0 $\mu\text{S}/\text{cm}$); Conductivity standard: (100,000 $\mu\text{S}/\text{cm}$)	Assess relative to the salinity standard.	$\pm 1\%$	1. Re-calibrate sensor 2. Call manufacturer 3. Send to factory for instrument servicing

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Sonde parameter	Calibration Frequency	Calibration Solutions	Calibration Verification	Acceptance Criteria	Corrective Action
DO	Daily	Outdoor calibration to 100% saturation of outdoor air	Assess 100% measurement relative to 100% outdoor air	± 0.2 mg/L	1. Re-calibrate sensor 2. Call manufacturer 3. Send to factory for instrument servicing
pH	Daily	4.0, 7.0, and 10.0 pH buffers solutions (NBS)	At the end of the sampling date (pH 7.0)	< 20 % of instrument drift	1. Re-calibrate sensor 2. Call manufacturer 3. Send to factory for instrument servicing

If the daily check indicates that the instrument's calibration has drifted outside the calibration acceptance criteria, the data will be flagged and evaluated following the procedures in Section B2 of this QAPP and an additional full calibration will be performed.

Secchi Depth

Water transparency will be measured with a 20 cm Secchi disk (black and white bands) attached to a 20 meters rope. Monitors will ensure that the disk is free of material and that the black/white stripes are completely visible and scratch- free.

B7. LABORATORY INSTRUMENT CALIBRATION, INSPECTION AND MAINTENANCE

The laboratory instruments or materials to be used for the analytical test methods will be calibrated, inspected and maintenance in accordance with the manufacturer recommendations and users guides or manuals, and based upon the test methods being used for the analysis (See Table 7).

Table 7: Laboratory Instrument or Materials Calibration and Inspection Schedule.

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Laboratory Instrument or Material	Test Parameter	Calibration Frequency	Inspection	Manufacturer Reference Guide Manual/Method
Thermo Scientific Orion Star™ A211 Benchtop Meter	DO/BOD	As per manufacturer specifications and recommendations.	The screw-on membrane with electrolytes will be visually inspected each day and replaced every 3 months according to manufacturer guidelines	Thermo Scientific Orion Star A-210 Series Electrochemistry Benchtop Meters Manual – June 2015.
Thermo Scientific GENESYS 50 UV-Vis Spectrophotometer	Chlorophyll-a	A certified chlorophyll-a standard solution will be analyzed each day of analysis as part of the QA/QC process.	Inspected and analyzed for any damage prior to daily analyses on the, which will be operated and maintained following manufacturer specifications	AquaMate 8100 UV-VIS Manual. SM10200 H
Enterolert Quanti-Trays	Enterococcus	N/A	Quality control testing on each new lot of Enterolert, sample containers and Quanti-trays with a positive control and a negative control.	SM 9230
Thermo Scientific Orion AQ4500 Turbidimeter	Turbidity	As per manufacturer specifications and recommendations. Calibrations and calibration check	The glass vials will be inspected prior to analysis of each sample to ensure they remain free of scratches and debris.	Thermo Scientific Orion AQUAfast AQ4500 Turbidimeter User Guide (2009). EPA 180.1.

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Laboratory Instrument or Material	Test Parameter	Calibration Frequency	Inspection	Manufacturer Reference Guide Manual/Method
		standards will be performed.		
Imhoff Cone	Settleable Solids:	N/A	Will be assessed for any visible damage prior to each measurement.	SM2540 F
Mettler Toledo ME204T Balance	TSS	Daily calibration and verification of calibration, prior to commencing work.	As per the manufacturer recommendations.	SM 2540 D
SEAL Analytical AA500 Atomic Absorption Continuous Flow Analyzer	TP, TKN, Nitrate + Nitrite	As per manufacturer specifications and recommendations. Calibrations and calibration check standards will be performed. All measurements will include daily calibration using certified materials	As per the manufacturer recommendations	Seal Analytical AA500 High Performance, High Automation Autoanalyzer Manual (2011). EPA 365.14 (TP), EPA 351.24 (TKN), and EPA 353.2 (Nitrate + Nitrite)
Apollo AS-ALK5	TA	N/A	Maintained and operated following manufacturer recommendations.	TA will be quantified relative to certified reference materials provided by the laboratory of Professor Andrew Dickson following best practices (Dickson et al. 2017). Field

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Laboratory Instrument or Material	Test Parameter	Calibration Frequency	Inspection	Manufacturer Reference Guide Manual/Method
				duplicate samples will be included to ensure repeatability between samples and adequate sample preservation.
Apollo AS-CS5	DIC	N/A	Maintained and operated following manufacturer recommendations.	DIC will be quantified relative to certified reference materials provided by the laboratory of Professor Andrew Dickson following best practices (Dickson et al., 2007).

B8. INSPECTION/ACCEPTANCE REQUIREMENTS FOR SUPPLIES AND CONSUMABLES

The CCRI chemistry lab manager will regularly assess all supplies and consumables to ensure proper working condition. All samples undergo rigorous checks for precision and accuracy to traceable standards as an additional check for proper functioning of instruments in the laboratory and quality assurance and quality control of data outputs for the measured parameters.

B9. NON-DIRECT DATA MEASUREMENTS & USE OF SECONDARY DATA

Non-direct measurements (also referred to as secondary data) are data that were previously collected under many different efforts outside this project. CCRI will use secondary data in the form of biological and physical monitoring data from coral reefs of Puerto Rico from the Puerto Rico Coral Reef Monitoring Program (PRCRMP). Because CCRI was not involved in the original data collection, our ability to detect errors will necessarily be limited to inconsistencies within the database, and to deficiencies described in the metadata. However, CCRI will perform basic quality assessment on numeric data provided to CCRI, including checks for anomalous values, extreme outliers, reported units, and excessive numbers of missing values (occasional missing values are

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not a concern for the large, long-term databases used in this project). CCRI will examine the performance criteria and quality objectives of the numeric data, if available, but those are often not reported.

The project will use the PRCRMP benthic and fish data, combined with the newly collected DEMO data, to calculate the BCG for each PRCRMP station. The BCG, initially developed by freshwater scientists, has been applied to the Caribbean coral reef ecosystem (Bradley et al. 2020a, b; EPA 2021; Santavy et al. 2021, 2022). The conceptual BCG describes how biological attributes of aquatic ecosystems change along a gradient of increasing human disturbance. The BCG Levels are biologically recognizable, measurable stages in the condition of coral reef ecosystems in response to increasing amounts of anthropogenic stress. The conceptual model has been calibrated for application to the near shore coral reefs of the USVI and Puerto Rico. The model can be used to support biological assessments of reef condition, monitor for changes in condition, identify high quality reefs, evaluate Best Management Practices (BMP) effectiveness, and support biological criteria development. Coupled with physical and chemical water quality data, the BCG forms a scientifically defensible basis to prioritize, protect, and restore water bodies containing coral reefs. The benthic BCG focuses on the structural and functional importance of benthic organisms including reef-building corals, algae, and other sessile invertebrates, how they interact, and how they indicate overall reef condition. The combination of the PRCRMP data with the DEMO data will allow us to calculate a BCG score for each PRCRMP station. On the other hand, the fish BCG focuses on the structural and functional roles of fish species and how they indicate reef condition.

Because fish and benthic communities respond differently to stressors, they can be combined for a robust assessment of biological condition. Calculating the Fish BCG Rules may be challenging, since PRCRMP only captures size data for fishes of commercial value (snappers, groupers, hogfishes, barracuda, jacks, mackerels, sharks) and/or fish species considered to be important reef herbivores (parrotfishes, doctor fishes). Therefore, the biomass calculations would be limited to those species and the biomass rule may be revised, based upon consultation with the BCG fish experts. BCG analyses will be done twice (1 per year) after conclusion of the PRCRMP and DEMO estimations.

B10. DATA MANAGEMENT

At each point where data and information are generated, compiled, or stored, the CCRI team must manage the information. Thus, the Information Management (IM) system includes all of the data-generating activities, all of the means of recording and storing information, and all of the processes which use data. The IM system also includes both hardcopy and electronic means of generating, storing, organizing and archiving data and the efforts to achieve a functional IM process is all encompassing. *To that end, all participants in the CCRI surveys play an integral part within the IM system.* **Table 8** provides a summary of the IM responsibilities identified by the CCRI group.

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Table 8: Summary of CCRI Information Management (IM) Responsibilities.

CCRI Group	Responsible Person	Primary Role	Responsibility
Field Teams	TBD	Acquire in-situ measurements and prescribed list of biotic/abiotic samples at each site targeted for the survey	• Complete and review field data forms and sample tracking forms for accuracy, completeness, and legibility. • Return field and sample tracking forms to CCRI so information can be integrated into the central database • Work with the CCRI staff to develop acceptable file structures and electronic data transfer protocols should there be a need to transfer and integrate data into the central database • Provide all data as specified in Field Operations Manual (EPA, 2010A) or as negotiated with the CCRI Project Leader. • Maintain open communications with CCRI staff regarding any data issues
Analytical Laboratories	TBD	Analyze samples received from field teams in the manner appropriate to acquire biotic/abiotic indicators/measurements requested.	• Review all electronic data transmittal files for completeness and accuracy (as identified in the Quality Assurance Project Plan). • Work with the CCRI staff to develop file structures and electronic data transfer protocols for electronic-based data. • Submit completed sample tracking forms to CCRI so information can be updated in the central database • Provide all data and metadata as specified in the laboratory transmittal guidance section of the Quality Assurance Project Plan or as negotiated with the CCRI Project Leader. • Maintain open communications with CCRI regarding any data issues.

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CCRI Group	Responsible Person	Primary Role	Responsibility
CCRI project manager	TBD	Provides support and guidance for all IM operations related to maintaining a central data management system for CCRI.	• Develop/update field data forms. • Plan and implement electronic data flow and management processes. • Manage the centralized database and implement related administration duties. • Receive, scan, and conduct error checking of field data forms. • Monitor and track samples from field collection, through shipment to the laboratory. • Receive data submission packages (analytical results and metadata) from the laboratory. • Run automated error checking, e.g., formatting differences, field edits, range checks, logic checks, etc. • Receive verified, validated, and final indicator data files (including record changes and reason for change) from QA reviewers. Maintain history of all changes to data records from inception through delivery to EPA's WQX. • Organize data in preparation for data verification and validation analysis and public dissemination. • Implement backup and recovery support for the central database. • Implement data version control as appropriate.
CCRI Project Manager	TBD	Review and evaluate the relevancy and quality of information/data collected and generated through the CCRI surveys.	• Monitor instrument and analytical quality control information. • Evaluate results stemming from field and laboratory audits. • Investigate and take corrective action, as necessary, to mitigate any data quality issues.

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CCRI Group	Responsible Person	Primary Role	Responsibility
CCRI Data Analysis and Reporting Team	TBD	Provide the data analysis and technical expertise needed for CCRI reporting requirements.	• Provide data integration, aggregation and transformation support as needed for data analysis. • Provide supporting information necessary to create metadata. Investigate and follow-up on data analysis activities that have been identified as data anomalies. • Produce estimates of extent and ecological condition of the target population of the resource. • Provide written background information and data analysis interpretation for report(s). • Document in-depth data analysis procedures used. • Provide mapping/graphical support. • Document formatting and version control.
Data Finalization Team	TBD	Provides data librarian support	• Prepare CCRI data for transfer to EPA WQX. • Generate data inventory catalog record (Science Inventory Record) • Ensure all metadata is consistent, complete, and compliant with EPA standards.
Abbreviations: TBD = To be determined.			

Data Flow Conceptual Model

CCRI will accumulate large quantities of observational and laboratory analysis data. To appropriately manage this information, it is essential to have a well-defined data flow model and documented approach for acquiring, storing, and summarizing the data. This helps focus efforts on maintaining organizational and custodial integrity, ensuring that data available for analyses are of the highest possible quality. Briefly, all field and lab sample analysis sheets will be stored on the instrumentation, as paper copies, and as digital backups. The finalized QA/QC sample data for each visit to each field site will be summarized into a .csv file and uploaded to appropriate data release for public viewing on CCRI and subsequent development into quarterly reports.

Simplified Data Flow Description

There are several components associated with the flow of information, these are:

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- Communication—between CCRI and the various data contributors (e.g., field crews, laboratories and the data analysis and reporting team)—is vital for maintaining an organized, timely, and successful flow of information and data.
- Data are *captured* or acquired from four basic sources —field data transcription, laboratory analysis reporting, automated data capture, and submission of external data files (e.g., GIS data)—encompassing an array of data types: site characterization; biotic assessment; and water quality analysis. Data capture generally relies on the transference of electronic data, e.g., optical character readers and email, to a central data repository. However, some data must be transcribed by hand in order to complete a record.
- Data repository or *storage*—provides the computing platform where raw data are archived, partially processed data are staged, and the “final” data, assimilated into a final, user-ready data file structure, are stored. The raw data archive is maintained in a manner consistent for providing an audit trail of all incoming records. The staging area provides CCRI a platform for running the data through all of its QA/QC paces as well as providing data analysts a first look at the incoming data. This area of the data system evolves as new data are gathered and user-requirements are updated. The final data format becomes the primary source for all statistical analysis and data distribution.
- Metadata—a descriptive document that contains information compliant with the Content Standards for Digital Geospatial Metadata (CSDGM) developed by the Federal Geographic Data Committee (FGDC).

Data Formats

Type of Data	Format
<i>Attribute Data</i>	Ascii Files: Comma-Separated values, or space-delimited, or fixed column
<i>GIS Data</i>	• ARC/INFO native and export files; compressed .tar file of ARC/INFO workspace • Spatial Data Transfer Standard (SDTS; FGDC 1999) format available on request

Standard Coding Systems

Information	Coding System
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<i>Sampling Site</i>	Latitude and Longitude in decimal degrees (+/- 7.4) Negative longitude values (west of the prime meridian). Datum used must be specified (e.g., NAD83, NAD27); EPA Locational Data Policy, EPA 1991
<i>Chemical Compounds</i>	CAS Number; Chemical Abstracts Service (CAS 1999)
<i>Species Codes</i>	Integrated Taxonomic Information System (ITIS 1999)
<i>Land cover/land use codes</i>	Multi-Resolution Land Characteristics (MRLC 1999); National Hydrography Dataset Plus Version 1.0 (NHDPlus 2005)

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Data Transfer Protocols

Field crews are expected to send in hard copies of field forms containing *in-situ* measurement and event information to CCRI as defined in this QAPP. Electronic data files are submitted by laboratories (and possibly some field crews). Field crews and labs must submit all sample tracking and analytical results data to the CCRI in electronic form using a standard software package to export and format data. Examples of software and the associated formats are:

<i>Software</i>	Microsoft Excel [®] Microsoft Access [®] SAS [®] R ArcMap [®]
<i>Export Options (file extensions)</i>	xls, xlsx, csv, formatted txt mdb, csv, formatted txt sas7bdat, csv, formatted txt csv, formatted txt

All electronic files must be accompanied by appropriate documentation, e.g., metadata, laboratory reports, QA/QC data and review results). This information should contain sufficient information to identify field contents, field formats, qualifier codes, etc. It is very important to keep CCRI informed of the completeness of the analyses. Labs may send files periodically, before all samples are analyzed, but CCRI must be informed that more data are pending if a partial file is submitted. All data files sent by the lab must be accompanied by text documentation describing the status of the analyses, any QA/QC problems encountered during processing, and any other information pertaining to the quality of the data. Following is a list of general transmittal requirements the laboratory should consider when packaging data for electronic transfer to CCRI:

- Provide data in row/column data file/table structure. Further considerations:
 - o Include sample id provided on the sample container label in a field for each record (row) to ensure that each data file/table record can be related to a site visit.
 - o Use a consistent set of column labels.

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- Use file structures consistently.
- Use a consistent set of data qualifiers.
- Use a consistent set of units.
- Include method detection limit (MDL) as part of each result record.
- Include reporting limit (RL) as part of each result record.
- Provide a description of each result /QC/QA qualifier.
- Provide results/measurements/MDL/RL in numeric form.
- Maintain result qualifiers, e.g., <, ND, in a separate column.
- Use a separate column to identify record-type. For example, if QA or QC data are included in a data file, there should be a column that allows the
- CCRI staff to readily identify the different result types.
 - Include laboratory sample identifier.
 - Include batch numbers/information or results can be paired with appropriate
- QA/QC information.
 - Include “True Value” concentrations, if appropriate, in QA/QC records.
 - Include a short description of preparation and analytical methods used to (where appropriate) either as part of the record or as a separate description for the test(s) performed on the sample. For example, EPAxxxx.x, ASTMxxx.x, etc. Provide a broader description, e.g., citation, if a non- standard method is used.

Include a short description of instrumentation used to acquire the test result (where appropriate). This may be reported either as part of the record or as a separate description for each test performed on the sample.

- Ensure that data ready for transfer to CCRI are verified and validated, and results are qualified to the extent possible.
- Data results must complement expectations (analysis results) as specified by contract or agreement.
- Identify and qualify missing data (why is the data missing).
- Submit any other associated quality assurance assessments and relevant data related to laboratory results (i.e., chemistry, nutrients). Examples include summaries of QC sample analyses (blanks, duplicates, check standards, matrix spikes) standard or certified reference materials, etc.), results for external performance evaluation or proficiency testing samples, and any internal consistency checks conducted by the laboratory.
- Labs may send electronic files by e-mail attachments, or they may upload files through a secure FTP location.

Data Quality and Results Validation

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Data quality is integrated throughout the life cycle of the data. Data received into CCRI from participants are examined for completeness, format compatibility, and internal consistency. Field collected data quality is evaluated using a variety of automated and other techniques. Analytical results are reviewed by subject matter experts. Any changes (deletions, additions, corrections) are submitted to the CCRI data center for inclusion into the validated data repository. All explanations for data changes are included in the record history.

Data Entry, Scanned, or Transferred Data

Field crews will record sampling event observational data in a standard and consistent manner using field data collection forms (**Appendices C, D and F**).

The CCRI staff either optically scans or transcribes information from field collection forms into an electronic format (sometimes using a combination of both processes). During the scanning process, incoming data are subjected to several automated error checking routines. Obvious errors are corrected immediately. Suspected errors that cannot be confirmed at the time of scanning are qualified for later review by someone with the appropriate background and experience (e.g., a chemist or aquatic ecologist). The process continues until the transcribed data are 100 % verified or no corrections are required.

Additional validation is accomplished by CCRI staff using a specific set of guidelines and executing a series of programs (computer code) to check for: correct file structure and variable naming and formats, outliers, missing data, typographical errors and illogical or inconsistent data based on expected relationships to other variables. Data that fail any check routine are identified in an “exception report” that is reviewed by an appropriate scientist for resolution.

CCRI staff brings any remaining questionable data to the attention of the CCRI director and individuals responsible for collecting the data for resolution.

Analytical Results Validation

All data are evaluated to determine completeness and validity. Additionally, the data are run through a rigorous inspection using SQL queries or other computer programs such as SAS or R to check for anomalous data values that are especially large or small, or are noteworthy in other ways. Focus is on rare, extreme values since outliers may affect statistical quantities such as averages and standard deviations. However, statistical outliers will not be removed from any data series and may only be flagged as questionable unless there is verifiable proof that a value was typed incorrectly or there was an error in the analysis.

All laboratory quality assurance (QA) information is examined to determine if the laboratory met the predefined data quality objectives - available through the Quality Assurance Project Plan (QAPP). In addition, DQAV will be performed to review the adherence to field and laboratory procedures in order to assess its quality and verify whether or not the data is reliable, scientifically valid, of known precision and accuracy, and of acceptable completeness and representativeness.

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If the data generated do not meet the quality objectives of this project, it will be flagged and reviewed to determine if appropriate quality controls are in place. Deviations from the analytical scheme described in this QAPP will be noted in the laboratory records associated with analytical batches in the QA statements, and the appropriated corrective actions will be implemented.

All questionable data should be corrected or qualified through the CCRI staff with support of the project manager and director.

Database Changes

Data corrections are completed at the lowest level by CCRI staff to ensure that any subsequent updates will contain only the most correct data. Laboratory results found to be in error are sent back to the originator (lab) for correction by CCRI staff. After the originator makes any corrections, the entire batch or file is resubmitted to CCRI. CCRI uses these resubmissions to replace any previous versions of the same data.

CCRI uses a version control methodology when receiving files. Incoming data are not always immediately transportable into a format compatible with the desired file structures. When these situations occur, CCRI staff creates a copy of the original data file which then becomes the working file in which any formatting changes will take place. The original raw data will remain unchanged. This practice further ensures the integrity of the data and provides an additional data recovery avenue, should the need arise.

All significant changes are documented by CCRI staff. CCRI includes this information in the final summary documentation for the database (metadata).

After corrections have been applied to the data, CCRI will rerun the validation programs to re-inspect the data.

Metadata

Computing Infrastructure

Electronic data are collected and maintained within a central server housed at the CCRI. Official IM functions are conducted in a centralized environment.

Data Security and Accessibility

CCRI ensures that all data files in the IM system are protected from corruption by computer viruses, unauthorized access, and hardware and software failures. Guidance and policy documents of EPA and management policies established by the IM Technical Coordination Group for data access and data confidentiality are followed. Raw and verified data files are accessible only to the CCRI collaborators. Validated data files are accessible only to users specifically authorized by the CCRI Project Leader. Data files in the central repository used for access and dissemination are marked as read-only to prevent corruption by inadvertent editing, additions, or deletions.

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Data generated, processed, and incorporated into the IM system are routinely stored as well as archived on redundant systems by the CCRI team. This ensures that if one system is destroyed or incapacitated, CCRI staff can reconstruct the databases.

Data security and accessibility standards implemented meet EPA's standard security authentication (i.e., username, password) process in accordance with the EPA's *Information Management Security Manual* (1999; EPA Directive 2195 A1) and EPA Order 2195.1 A4 (2001D). Any data sharing requiring file transfer protocol (FTP) or internet protocol is provided through an authenticated site.

Life Cycle

Data may be retrieved electronically by the CCRI team, partners and others throughout the records retention and disposition lifecycle or as practicable.

Data Recovery and Emergency Backup Procedures

CCRI maintains several backup copies of all data files and of the programs used for processing the data. Backups of the entire system are maintained off-site by CCRI staff. The IM process used by CCRI, uses system backup procedures. The CCRI team backs up and archives the central database according to procedures already established for UPRM. All laboratories generating data and developing data files are expected to establish procedures for backing up and archiving computerized data.

Long-Term Data Accessibility and Archive

All data are transferred by CCRI's staff to U.S. EPA's agency wide WQX data management system for archival purposes. WQX is a repository for water quality, biological, and physical data and is used by state environmental agencies, EPA and other federal agencies, universities, private citizens, and many others. Revised from STORET, WQX provides a centralized system for storage of physical, chemical, and biological data and associated analytical tools for data analysis. Data from the CCRI project in an Excel format will be run through an Interface Module and uploaded to WQX by CCRI staff. Once uploaded, states and tribes and the public will be able to download data (using Oracle software) from their region.

Records Management

Removable storage media (i.e., CDs, diskettes, tapes, flash storage, hard drives) and paper records are maintained in a centrally located area at CCRI. Paper records will be returned to the CCRI project manager once the assessment is complete. CCRI staff identifies and maintains files using standard divisional procedures. Records retention and disposition comply with U.S. EPA directive 2160 Records Management Manual (July 1984) in accordance with the Federal Records Act of 1950.

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SECTION C: ASSESSMENT/OVERSIGHT

C1. CORRECTIVE ACTIONS

Any duplicate samples that exceed the thresholds for precision or exceed limits of instrument drift for repeated calibration samples will be flagged accordingly. Field samples (i.e., T, S, pH, DO, Secchi depth) will be repeated on site for any flagged samples. Lab samples will be re-analyzed if possible and or flagged in the final reports with a memorandum or corrective action notice.

C2. REPORTS TO MANAGEMENT

CCRI will be delivering two types of reports and additional deliverables. The reports will be submitted electronically as pdf documents and consist of:

1. Quarterly reports, which will be produced after a sampling cycle is concluded (for a total of 8 reports); these reports will include a detailed description of the activities performed during that sampling cycle, supported by visual documentation such as photographs or videos of the monitoring events. These reports will also contain all the raw data collected in each cycle, estimations of water quality parameters based on that raw data, and the final spreadsheets that will be uploaded in the Water Quality Portal (see above). Very important, the report will also include all the Data Quality Assurance/Quality Control (QA/QC) processes described in the QAPP and validation reports for the analytical results. Financial reports detailing costs incurred during the reporting period will also be included.
2. Final reports: CCRI will produce a final report that will contain:
 - a. Comprehensive datasets - e.g., these should include raw data as well as any calculations of desired parameters and associated metadata (Excel Spreadsheets).
 - b. Project water quality data results (Excel Spreadsheets used to upload the information to the Water Quality Data Portal).
 - c. Data Quality Assurance/Quality Control (QA/QC) and validation reports for the analytical results gathered in the water quality monitoring project.
 - d. General data analyses and interpretation on temporal and spatial trends.
 - e. Correlations with the PRCRMP benthic and fish data collected during the same 2-year period.

If appropriate, data and interpretations will eventually be published in peer-reviewed journal articles.

SECTION D: DATA VALIDATION AND USABILITY

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D1. DATA REVIEW, VALIDATION, AND VERIFICATION REQUIREMENTS

Any duplicate samples that exceed the thresholds for precision or exceed limits of instrument drift for repeated calibration samples will be flagged accordingly. Field samples (i.e., T, S, pH, DO, secchi depth) will be repeated on site for any flagged samples. Lab samples (i.e.,) will be re-analyzed if possible and or flagged in the final reports with a memorandum or corrective action notice.

Coral survey data will be entered into a standardized electronic data sheet developed in MS Excel®. Records will be entered by someone other than the surveyor and verified for accuracy by the surveyor within 24h. The database will be compiled and backed up on CD each evening in a separate file. The recorder will validate transcription of the data sheets to the electronic spreadsheet, and the P.I. will verify that this has been completed. Underwater data sheets will be retained by the Research Assistant 1, who will note any deviations from these procedures on the data sheets. Before analysis, a final check will be performed to avoid transcription errors; at least 10% of the data will be validated. If any corrections are required, both the original and a corrected copy of the electronic file will be retained. If data entry errors are found to exceed 1% of those checked, all data will be validated. Once data are checked for quality, they will be made available for analysis in a read-only format. Data management may be performed on Excel spreadsheets.

Raw photographic data will be transferred within a day to an electronic database by the photographic diver. Photographic images will be saved on a computer with an appropriate file name

D2. DATA VALIDATION AND VERIFICATION

All samples will be analyzed in accordance with standardized methods with respect to reference materials and replicate samples (Table 5). Any values that fall outside of these required test parameters will be flagged (see corrective actions below). The certified chemist and project PIs will be in charge of verifying and validating all data using graphical and or statistical approaches to flag any additional potential outlying data points. These secondary flagged data points will undergo the same corrective actions as defined in the subsequent sections. The main validation will be carried out by the Caribbean Coral Reef Institute. However, the final validation of the data is the responsibility of the office of certainty and quality control of the DNER, in accordance with the standardized procedures for field sampling and validation of laboratory data, the data quality objectives and the control limits.

D3. RECONCILIATION WITH DATA QUALITY OBJECTIVES

Data and results from each aspect of the study will be reviewed by the Research Assistants I and II and reconciliation with data quality objectives will proceed as required, including discussions with the data collectors. Methods used and results obtained will be discussed with other project members as warranted. QA/QC audits may be conducted at any time during the studies. Any limitations on use of the data will be conveyed to decision makers by qualifiers, footnotes, etc. in any reports emanating from the project.

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APPENDIX A: STONY CORAL DEMOGRAPHIC SURVEY

Stony coral demographic surveys document the taxa, 3D size, amount of tissue on coral colonies and the occurrence of adverse health conditions such as bleaching, disease or overgrowth by boring sponges. These characteristics can provide estimates of stony coral abundance, density, species diversity, richness, reef surface area and complexity, and relative health of coral species.

Surveyors must be able to identify corals to genus and species, measure coral colony dimensions and estimate the proportion of tissue in relation to the overall size of the colony. Surveyors must note any bleaching, disease, and invasive growth of clonid sponges, taking care to distinguish from predation damage.

Data collection occurs in three (3) phases: (1) Predive, (2) Coral Demographic and (3) site assessments.

1. Predive

Station information is to be recorded in two primary data logs prior to entering the water: **Boat/Dive log** (Figure A1) and **Datasheet** (Figure A2).

Boat Log. The Boat Driver completes the Boat Log. There are two divers at each site, the DEMO diver (lead), who collects all data, and the Buddy diver (Buddy), who photographs the site and ensures safety protocols are met. The names and roles of both divers are recorded on all data sheets associated with the dive.

Date	DO D	Site	Role	Diver	O2%	PSI In	Time In	Time out	Depth
8/12/22	1	12345678	DEMO	Smith					
8/12/22	1	12345678	Buddy	Jones					
8/12/22	2	98765432	DEMO	Smith					
8/12/22	2	98765432	Buddy	Jones					

Figure A1. Example of boat log with station information filled out. The first dive of the day consisted of two divers, one DEMO and one photographer/videographer. The DEMO diver is Smith, and the Buddy diver is Jones. DOD = Dive of the day; site = PRCRMP site ID; Role = role assigned to each diver. O2% refers to the oxygen blend in the tanks, PSI is the amount of air in each tank, and time in is when the divers enter the water

Datasheet. Data collected are consistent with the NCRMP programmatic methodology. Logistic and station information – Names of all divers, Field ID, date, time of survey, and mission data manager are completed prior to divers entering the water (Figure A2). Fill in all categories legibly.

Field ID – The **Field ID** is a unique alpha-numeric number the diver is to record on the datasheet at each station.

FIELD ID = (SITE #) + (STATION #)

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Date	R Locality	Transect
Diver	Buddy	depth
Observed Habitat Type		

Figure A2. Coral Demographic datasheet header with logistic and station information.

2. Coral Demographic

The following field equipment is required:

- Hand-held GPS with the positions of localities and transects within each locality
- Demographics datasheet, clipboard, pencil, spare pencil ○ One survey per datasheet
- Small rigid measuring instrument, marked in cm (e.g., “Flexiruler”)
- Measuring instrument, one meter long marked in cm increments, used for measuring coral colony dimensions AND/OR for measuring 1m out from the transect tape (e.g., 1m PVC, marked in units or with measuring tape securely attached)
- Camera, battery, housing
- Video camera, battery, housing

Navigating to the site. Each day, dive boats will be deployed from shore to carry divers and equipment to the designated PRCRMP sites. Once in the field, the boat captain navigates to the selected PRCRMP site using a handheld GPS unit. Boat operators will anchor (avoiding corals) near the site using hand-held GPS devices to locate and document latitude and longitude coordinates.

The DEMO team enters the water at selected GPS coordinates, descends to bottom, locates the permanently marked transects and affixes the surface float line to the bottom.

Establishing transect. The demographic survey locality has five, 10m long permanent transects. Each one marked by two re-bars and a code number with cable-ties. The survey area is 10m long by 1m wide along each permanent transect). The DEMO team will tie the transect tape to the re-bar with the cable ties. Roll out the tape keeping it taut 10m to the next re-bar marker using weights clipped to the transect tape. The DEMO diver will avoid wrapping the tape around substrate or biotic object, as this will distort sampling distances and locations for the benthic divers. The end of the tape should be tied or clipped to the re-bar so that the transect tape is as taut as possible. If site is pavement or scattered coral in sand, soft weights may be used to weight the transect tape at the beginning and end to keep transect in place. The demographic survey is conducted **ONLY** along the **LEFT** side of the transect line (tape), from the origin (meter marker zero). This means the area of the survey is to the left side of the tape. Do not split the survey area 1/2 a meter on either side of the transect tape.

IMPORTANT: A new datasheet is to be used for each demographic survey, one survey per sheet (i.e., do not record data for survey Y on the back of survey X’s datasheet). ***This is critical for data management.***

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Observed habitat type. Identification of the habitat type (see Appendix B) observed at the diver scale. At the surface, the diver should discuss with their buddy and other team members and try to come to a consensus. If a consensus is not achieved divers should note that in the Field/Boat Log.

Coral Demographic Survey Protocols

The DEMO diver collects the species/colony information within each 10x1m band transect: Identification and additional colony information of all visible scleractinian corals within the survey area (left 10 x 1m = 10 sqm of transect) will be recorded.

The Demographic diver collects the following information within the 10x1m band transect:

Species/colony identification – Each individual scleractinian coral colony with all or any part of the living colony or skeletal unit within the transect area will be identified to species level (Figure A3). **Note: Appendix B** provides the list of coral species and the few octocoral that will be ID'd to species level using a simplified, universally used code of four letters, a high cap letter for the genus and three low-cap letters for the species (*Orbicella faveolata* = Ofav).

- Record each individual colony on datasheet.
- All visible corals of any size will be identified to the species level (**Figure A3, Appendix B**). If species-level identification is not possible, take photo for later analysis.
- When a scleractinian colony smaller than 5cm maximum dimension (< 5cm) of a species is encountered in the transect area (**Figure A3**):
 - Identify the coral to lowest possible taxonomic resolution and record on datasheet.
 - Specific size measurements and colony condition (bleached and diseased information) are not recorded for juveniles. Draw a line through the rest of the row and continue to the next coral. Count all juveniles.
- When a scleractinian colony with a maximum skeletal dimension of greater than 5 cm (>5cm) is encountered in the transect area, continue with ALL of the following measurements (Fig. A4).
- .

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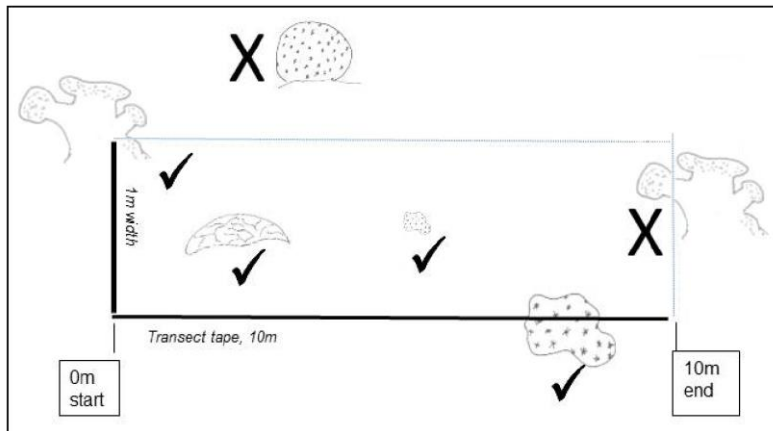


Figure A3. Schematic of example 10m x 1m transect area. Corals with all or part of colony (excluding branches) within transect area are included. Corals entirely outside of the transect area are not included (X).

Live percent coral tissue

Coral colony size measurements - Measure entire coral (skeleton + live tissue) on a planar dimension (2D) to three (3) exact dimensions (cm). If coral colony is partially in the transect, measure the entire colony AND measure the portion in the transect (two measurements)

- Measurements made to the nearest whole centimeter (cm).
- Do not bin, estimate, or aggregate measurements. For example, measurements of length, width, and height of a colony might be 5cm x 3cm x 2cm, respectively. Those that are crust-like with no height, will have only two measurements (Fig. A5-A).
 - Maximum diameter Length – Measure the maximum diameter (cm) of identifiable skeletal unit (**Figures A4 -A and B**).
 - Measure location where diameter of skeletal unit is widest
 - Measure whole colony-skeletal unit, not just the live tissue
 - Maximum perpendicular diameter width – Measure the perpendicular diameter of skeletal unit at its' greatest width (**Figure A4-A**). Maximum diameter length is to be greater than perpendicular diameter width.

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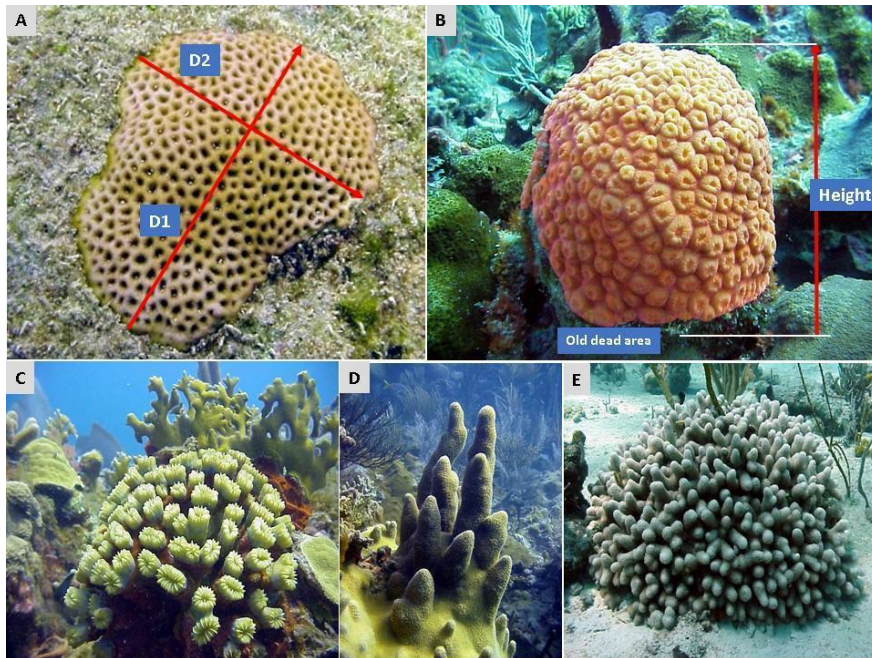


Figure A.4: Colony size measurements. Maximum (D1) and perpendicular diameter (D2) (A), and height (B). Photos of branching *Eusmilia fastigata*, columnar *Dendrogyra cylindrus* and branching *Porites porites*. All three have skeletal continuity (a single genet) even though, in *E. fastigata* and *P. porites* colonies, there is no physiological (tissue) continuity.

- a. Height – Measure the height (cm) of the skeletal unit (**Figure A4-B**).
- b. Height is measured from the base of the skeletal unit perpendicular to plane of growth.
 - i. If colony is growing on a slope, measure perpendicular to the slope (**Fig. A5**)
 - ii. Measure linearly (i.e., do not drape tape across the colony)
 - iii. If the colony has an encrusting morphology, the minimum height of the colony should be reported as 1.0 cm.

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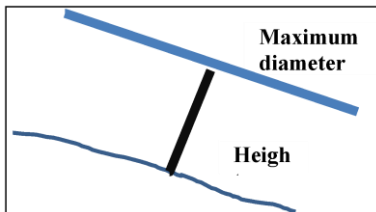


Figure A5. Height and width measurements perpendicular to plane of growth. Colony is shown on a sloped reef.

- Thickets/clumps. If the skeletal unit is connected, identify as one individual. If not, then record them as multiple individuals. Species such as *Acropora cervicornis*, *Acropora palmata*, *Eusmilia fastigiata*, *Porites porites*, *P. furcata*, *Madracis auretenra*, *Orbicella annularis* and *Dendrogyra cylindrus* may have large, single genotype colonies that suffered partial mortality and therefore, left ramets (surviving areas) that are not physiologically connected, that are separated by tissue-devoid skeleton covered by turf algae, for example. Individual colonies in dense thickets of acroporid or branching *Porites* are usually impossible to separate. In cases like this, the area of the live thicket, if continuous should be recorded as one colony and a note made in the “observation” slot in the data sheet
3. **Coral condition**– For each measured coral along the band transect, the total colony area (3D) is assessed for healthy live tissue, biotic disease, bleaching and mortality. NOTE: coral condition is not assessed for juvenile colonies, but juvenile colonies should be counted.
- i. Bleached (BLE): most of the colony shows signs of bleaching. Bleached colony is one with a pale to bright white coloration that still has live tissue, but it is transparent due to the loss of zooxanthellae, the micro algae that provides coloration. Presence of live tissue must be checked by touching the white areas. It will feel mucous.
 - ii. Partial bleaching (PBL): a section of the colony or spotted areas are white or pale.
 - iii. Paling (PAL): a condition where areas of coral tissue experience a loss of zooxanthellae in response to stress so that the spots or whole coral colony appears paler in color compared to normal coloration.
 - iv. No bleaching (HEA= healthy)): no bleaching or disease present, colony shows normal coloration. Use this code to indicate no bleaching. Do not leave this item blank.

¹ NOTE: *Siderastrea siderea* and *S. radians* usually appear bright blue rather than white when bleached.

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4. **Coral Disease** – Note if the coral being measured has signs of a disease and use the coral disease regular codes (Acronyms) - BBD= black band disease, WPD= white plague disease, SCTLD= stony coral tissue loss disease, CYBD= Caribbean yellow band disease, DSD= dark spots disease, etc.) (**Figure A6 and 9A**). Again, a healthy-looking colony is HEA. Coral disease prevalence will be recorded at the species level. Disease reports will be transcribed when back in the lab.

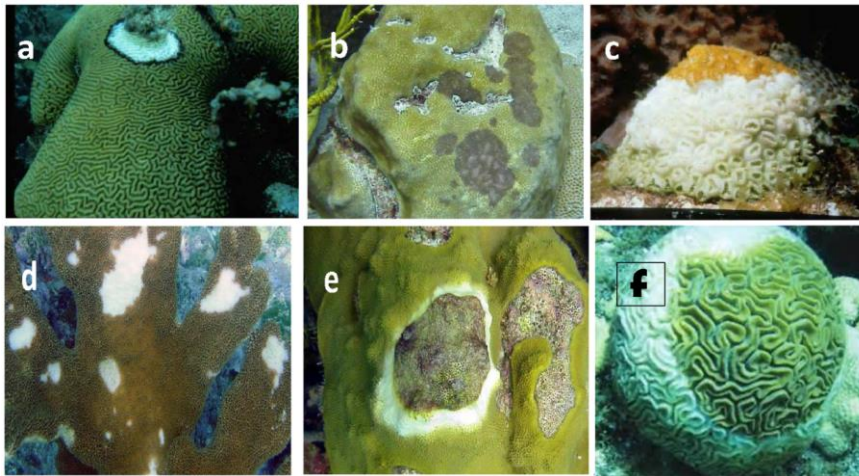


Figure A6-1. Examples of coral diseases: a) black band disease (BBD), (b) dark spots disease (DSD), c) white plague disease (WPD), d) white pox disease or , e) Caribbean yellow band disease (CYBD) and f) stony coral rapid tissue loss disease (SCTLD). In examples a, c, d and f, the disease is advancing fast, note the white band between live tissue and turf-colonized skeleton. In b, no signs of mortality are observed, and in e, the pale area around the dead center, is bleached tissue that dies slowly.

4. **Notes.** Additional notes may be added into the bottom of the datasheet.

- Used to indicate any field observation that may be important, such as presence of an ecological incident (e.g., *cliona* infection, injured (predation, other)).
- If data collection requires more than one datasheet (both sides), the diver will indicate the presence of multiple sheets in the NOTES section of the datasheet.
- Notes are entered into the database, but they are cataloged separately for reference.

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Appendix B: CCRI Species Codes

CCRI code	Species Name
Acer	<i>Acropora cervicornis</i>
Apal	<i>Acropora palmata</i>
Apro	<i>Acropora prolifera</i>
Ac-sp	<i>Acropora sp.</i>
Aaga	<i>Agaricia agaricites</i>
Afra	<i>Agaricia fragilis</i>
Ahum	<i>Agaricia humilis</i>
Alam	<i>Agaricia lamarcki</i>
Apus	<i>Agaricia pusilla</i>
Agsp	<i>Agaricia sp.</i>
Aund	<i>Agaricia undata</i>
Carb	<i>Cladocora arbuscula</i>
Cnat	<i>Colpophyllia natans</i>
Cama	<i>Colpophyllia amaranthus</i>
Cbre	<i>Colpophyllia breviserialis</i>
Dcyl	<i>Dendrogyra cylindrus</i>
Dsto	<i>Dichocoenia stokesii</i>
Dlab	<i>Diploria labyrinthiformis</i>

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CCRI code	Species Name
Disp	<i>Diploria spp</i>
Efast	<i>Eusmilia fastigiata</i>
Ffra	<i>Favia fragum</i>
Hcuc	<i>Heliocercis cucullata</i>
Irig	<i>Isophyllia rigida</i>
Isin	<i>Isophyllia sinuosa</i>
Issp	<i>Isophyllia spp</i>
Maur	<i>Madracis auretenra</i>
Mdec	<i>Madracis decactis</i>
Mfor	<i>Madracis formosa</i>
Mpha	<i>Madracis pharensis</i>
Msen	<i>Madracis senaria</i>
Ma-sp	<i>Madracis spp</i>
Maer	<i>Manicina areolata</i>
Mdan	<i>Meandrina danae</i>
Mjac	<i>Meandrina jacksoni</i>
Mmea	<i>Meandrina meandrites</i>
Me-sp	<i>Meandrina spp</i>

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CCRI code	Species Name
Mcav	<i>Montastraea cavernosa</i>
Mang	<i>Mussa angulosa</i>
Mali	<i>Mycetophyllia aliciae</i>
Mdan	<i>Mycetophyllia danaana</i>
Mfer	<i>Mycetophyllia ferox</i>
Mlam	<i>Mycetophyllia lamarckiana</i>
My-sp	<i>Mycetophyllia spp</i>
Odif	<i>Oculina diffusa</i>
Oc-sp	<i>Oculina spp</i>
Oval	<i>Oculina valenciennesi</i>
Orob	<i>Oculina robusta</i>
Oann	<i>Orbicella annularis</i>
Ospx	<i>Orbicella annularis species complex</i>
Ofav	<i>Orbicella faveolata</i>
Ofra	<i>Orbicella franksi</i>
Or-sp	<i>Orbicella spp</i>
Past	<i>Porites astreoides</i>
Pdiv	<i>Porites divaricata</i>

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CCRI code	Species Name
Pfur	<i>Porites furcata</i>
Ppor	<i>Porites porites</i>
Po-sp	<i>Porites spp</i>
Pcli	<i>Pseudodiploria clivosa</i>
Pstr	<i>Pseudodiploria strigosa</i>
Ps-sp	<i>Pseudodiploria spp</i>
Scub	<i>Scolymia cubensis</i>
Slac	<i>Scolymia lacera</i>
Sc-sp	<i>Scolymia spp</i>
Srad	<i>Siderastrea radians</i>
Ssid	<i>Siderastrea siderea</i>
Si-sp	<i>Siderastrea spp</i>
Sbou	<i>Solenastrea bournoni</i>
So-sp	<i>Solenastrea spp</i>
Sint	<i>Stephanocoenia intersepta</i>
Taur	<i>Tubastraea aurea</i>

Other important benthic species and groups

Common susceptible Octocorals	
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Gvent	<i>Gorgonia ventalina</i>
Gflab	<i>Gorgonia flabellum</i>
Gar	<i>Gorgonia mariae</i>
Ecar	<i>Erythropodium caribaeorum</i>
Basb	<i>Briareum asbestinum</i>
Pame	<i>Pseudoterogorgia americana</i>
Zoanthids	
Pcar	<i>Palithoa caribaeorum</i>
Zsoc	<i>Zoanthus sociatus</i>
Sponges	
Cten	<i>Cliona tennuis</i>
Avar	<i>Antosigmella varians</i>

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Appendix C: Example of CCRI boat/dive Log

Date:

Supervisor's Dive Log
of _____

Page

Date	DO D	Field ID	Station	Diver	O2%	PSI In	Time In	Actual Dive		Time Out	PSI Out	Comments
								Depth	BT			

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Appendix D: Coral Demographic Datasheet

[illegible]

D1= maximum diameter

D2= Perpendicular maximum diameter

Height

M1- maximum diameter dead area

M2= Perpendicular maximum diameter dead area

Health: healthy (H), diseased (WPD, SCTL, CYBD etc.),

Bleach = T, PB, P, N

JUV = number of juvenile

Notes = other observations (clona infection, injured (predation, other))

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Appendix E: Habitat Types

Aggregate reef: Continuous, high-relief coral formation of variable shapes. Examples of aggregate reefs include fore reef, fringing reef, shelf edge reef, spur and groove reef)

Bedrock: Exposed bedrock contiguous with the shoreline. May be colonized (dense coverage of macroalgae, hard coral, gorgonians, and other sessile invertebrates) or uncolonized (often covered by a thin sand veneer with sparse coverage of biota (macroalgae, hard coral, etc.).

Patch reef: Coral formations that are isolated from other coral reef formations by sand, seagrass or other habitats. Can be an individual patch reef or aggregate patch reefs.

Individual Patch Reef

Usually circular or oblong. High relief. Isolated by other formations by sand, seagrass. At the scale of the transect, may look like aggregate reef or even pavement.

Aggregate Patch Reef

Circular or oblong features not big enough to map individually. The cluster of hardbottom is >10% of the entire polygon. Where there is a halo, the halo is part of the polygon.

Pavement: Flat, low-relief, solid carbonate rock. May be colonized (dense coverage of macroalgae, hard coral, gorgonians, and other sessile invertebrates) or uncolonized (often covered by a thin sand veneer with sparse coverage of biota (macroalgae, hard coral, etc.).

Scattered coral/rock in sand: Primarily sand or seagrass bottom with scattered rocks or small, isolated coral heads that are too small to be delineated individually (i.e., smaller than individual patch reef).

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Appendix F: Example of Field data Sheet

Field Team Member Names: _____

Date								
Time								
Site Code								
Replicate	R1	R2	R1	R2	R1	R2	R1	R2
Surface Temperature (°C)								
Surface Salinity (psu)								
Surface DO (mg/L)								
Surface pH_NBS								
Secchi Depth 1 (m)								
Secchi Depth 2 (m)								
Secchi Depth 3 (m)								
Bottom Temperature (°C)								
Bottom Salinity (psu)								
Bottom DO (mg/L)								
Bottom pH_NBS								
BOD Bottle Number								
Chl-a Bottle Number								
Enterococcus Bottle Number								

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Turbidity Bottle Number								
Settleable Bottle Number								
TSS Bottle Number								
Nutrients Bottle Number								
TA/DIC Bottle Number								

Notes:

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Appendix G: CCRI Water Quality Monitoring Lab Data Management Plan

- **Field Collected Data:** All field data sheets will be photographed following each sampling day and archived onto computer and cloud-based storage. The physical data sheets will be digitized within 48 hours of sampling and verified for QA/QC of data input by a third party within CCRI. The physical data sheets will be placed in physical archives within the CCRI Water Quality Monitoring Lab and the verified, digitized data sheets will be archived onto computer and cloud-based storage. Field-based measurements (T, S, DO, pH, secchi depth) will be assessed for QA/QC using field-based replicates for desired precision (**Table 5**) and assessment of daily *Hydrolab* Calibration Sheets (**Appendix I**) before being compiled into the preliminary water quality data sheets. The field-based depth profiles (T, S, depth) from the *Hydrolab* are downloaded from the instrument following each sampling day and are stored into CCRI data archives.
- **Lab Collected Data:** All field-collected sample vials will be transported to the CCRI Water Quality Monitoring Lab (BOD, Chlorophyll-a, Enterococcus, Turbidity, Settleable Solids, Total Suspended Solids, Total Phosphorus, Total Kjeldahl Nitrogen, Nitrite + Nitrate) or UPRM Biogeochemistry and Ecology Research Group (TA, DIC) for analysis of the desired water quality parameters. Total Nitrogen will be determined as the sum of Total Kjeldahl Nitrogen + Nitrite + Nitrate (Puerto Rico Water Quality Standard Regulation, 2019). All samples will be analyzed according to EPA approved methods, including holding times (**Table 5**), and assessed for accuracy and precision following the respective method protocols for replicate samples, sample spikes, and certified reference materials (**Table 4**) to ensure that data quality assurance objectives are met. Independent paper lab notebooks and digitized spreadsheets will be kept for the raw instrument data output for each measurement. The resulting preliminary data will be compiled into the preliminary water quality data sheets.
- **Preliminary Water Quality Data Sheets:** The preliminary water quality data sheets will include columns for date, time, site code, depth, T, S, DO, pH, secchi depth, BOD, Chlorophyll-a, Enterococcus, Turbidity, Settleable Solids, Total Suspended Solids, Total Phosphorus, Total Kjeldahl Nitrogen, Nitrite + Nitrate, and Total Nitrogen. Independent measurements for each sampling event will be recorded as rows within this compiled data format. Missing or bad data points will be listed as -999 in the datasheet and data below the respective detection limit for the measurement will be listed as 0. The preliminary water quality data sheets will be QA/QC'd by the Chemistry Laboratory Manager and at least 10% of the observations will be QA/QC to match the originating data records by another person. The Chemistry laboratory manager will review all instrument protocols and certify the data in compliance with the aims and objectives of the water quality monitoring project (Puerto Rico Water Quality Standard Regulation, 2019). CCRI Researcher, Dr. Travis Courtney, will additionally review all data for QA/QC and flag any additional points for review before finalizing the data reports.
- **Finalized Water Quality Data Sheets:** All finalized data sheets will be archived on computers, hard drives, and cloud-based storage following the data management section of the

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QAPP. The finalized data sheets will additionally be appended to the finalized data sheet that is updated following each sampling and archived on the CCRI computers, hard drives, and cloud-based storage.

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Appendix H: Example of CCRI Water Quality Monitoring Chain of Custody Form

Field Team Leader Name: _____

Date/Time Sampled	Site Name	Site Code	BOD (Qty)	Chl-a (Qty)	Entero (Qty)	Turbidity (Qty)	TSS (Qty)	Settleable (Qty)	TKN (Qty)	N+N (Qty)	TA+DIC (Qty)

Signature of Field Team Leader Sampled According to SOP: _____

Samples and Field Data Sheets Received/Relinquished for delivery by:

Print Name	Signature	Date & Time Relinquished	Date & Time Received

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Samples and Field Data Sheets Received/Relinquished at Laboratory by:

Print Name	Signature	Date & Time Relinquished	Date & Time Received

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Appendix I: Hydrolab HL4 Calibration Sheet

Date: _____

Time: _____

Name of Calibrator: _____

Temperature Calibration:

Portable Meter = _____ NIST Thermometer = _____

pH_{NBS} Calibration:

7.00 = _____ 10.01 = _____

Dickson Tris Seawater pH_T Standard = _____

Conductivity Calibration:

100 μ S/cm = _____ 1413 μ S/cm = _____ 12.9 mS/cm = _____

36 ppt Salinity Standard = _____

Dissolved Oxygen Calibration:

Saturated Air = _____

Calibration Initials: T = _____ pH = _____ S = _____ DO = _____

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Appendix J: CCRI Water Quality Field Sampling Plan

Following navigation to the respective sampling station coordinates using GPS, the boat is kept approximately stationary at the site and the following samples are collected and recorded on the waterproof datasheets using #2 pencils. A second observer on the boat will be present to ensure that all handwriting is clear and legible at the time of data collection.

Surface, Bottom, and Depth Profiles for T, S, DO, pH: The Hydrolab HL4 probes are submerged directly into the surface water (~0.1 m depth) for the surface sample to record surface T, S, DO, and pH on the data sheets. The Hydrolab HL4 sonde is then lowered until it reaches approximately the bottom where T, S, DO, and pH are recorded on the data sheets. The instrument is then pulled back up into the boat. The instrument software automatically quantifies GPS, time, T, S, DO, and pH for each cast to upload the full surface-to-bottom depth profile of those parameters for each site visit. These are then recorded on the Field Data Sheet.

Secchi Depth: The secchi disk, which is a 20 cm black and white disk with non-stretchable line that is marked at 0.5 m intervals, is lowered from the boat until it is no longer visible by the observer. Two standardized, identical secchi disks will be on hand at CCRI in the event that one is lost or irreparably damaged to maintain consistency across samples. The disk is assessed from the shady side of the boat by an observer who is not wearing sunglasses/hats/or any other eye shades and ideally is the same observer across all samples and all sites. The depth at which the secchi disk disappears and reappears is quantified 3 times to the nearest 0.1 m interpolated from the 0.5 m depth intervals of the secchi disk line. If any of the measurements differ by more than 0.5 m the measurement is repeated. The Secchi depth is subsequently recorded in the Field Data Sheet.

Bottom Samples:

A 6L Van Dorn water sampler is lowered to approximately 0.5 m above the seafloor and the messenger is released to collect the bottom seawater samples. The total sample volume is ~4.5L, which leaves an additional 1.5 L available in the water sampler for field duplicates of selected parameters at each site. ~~A separate water sampler and water transfer tubing is used for bottom enterococcus samples to avoid contamination between sites and samples. The device is soaked in 10% bleach solution with a sodium thiosulphate rinse to neutralize the chlorine prior to each sampling day. For example, if 4 sites are visited in a day, then 4 clean water samplers will be available on the boat to sample the water at each of those 4 sites to avoid cross contamination of enterococcus counts between sites.~~ After each sample collection, the water sampler is assessed for leaks on the boat following best practices and sampled for the following parameters and preserved according to their respective methods and volumes (Tables ~~a~~ 5,8):

1. Seawater Total Alkalinity and Dissolved Inorganic Carbon ~~(60 mL)~~
2. Biochemical Oxygen Demand ~~(300 mL)~~

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3. **Chlorophyll-a** ~~(1 L)~~
4. **Enterococcus** ~~(250 mL)~~
5. **Turbidity** ~~(250 mL)~~
6. **Settleable Solids** ~~(1 L)~~
7. **Total Suspended Solids** ~~(1 L)~~
8. **Total Phosphorus and Total Kjeldahl Nitrogen** ~~(500 mL)~~
9. **Nitrite + Nitrate** ~~(125 mL)~~

All reusable equipment, including the Van Dorn Bottle, will be rinsed with potable water, scrubbed with phosphate-free soap and a clean brush to remove particulates, rinsed with potable water (5x), rinsed with distilled water (5x), air dried, and stored dried and sealed in a clean packaging to prevent any contamination prior to the next sampling.

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